Bundesinstitut für Risikobewertung

AI-based autonomous agents - What, why and how?

07.04.2025

Presenter: Dr. Iurii Savvateev

Outline

Part I. What are AI-based autonomous agents?

Break I

Part II. Why and in which areas to use agents?

Break II

Part III. How to practically start using agents?

Outline

Part I. What are AI-based autonomous agents?

Break I

Part II. Why and in which areas to use agents?

Break II

Part III. How to practically start using agents?

Part I. What?

We define AI as the study of agents that receive percepts from the environment and perform actions.

— *Artificial Intelligence: A Modern Approach*, Stuart Russell and Peter Norvig (2003).

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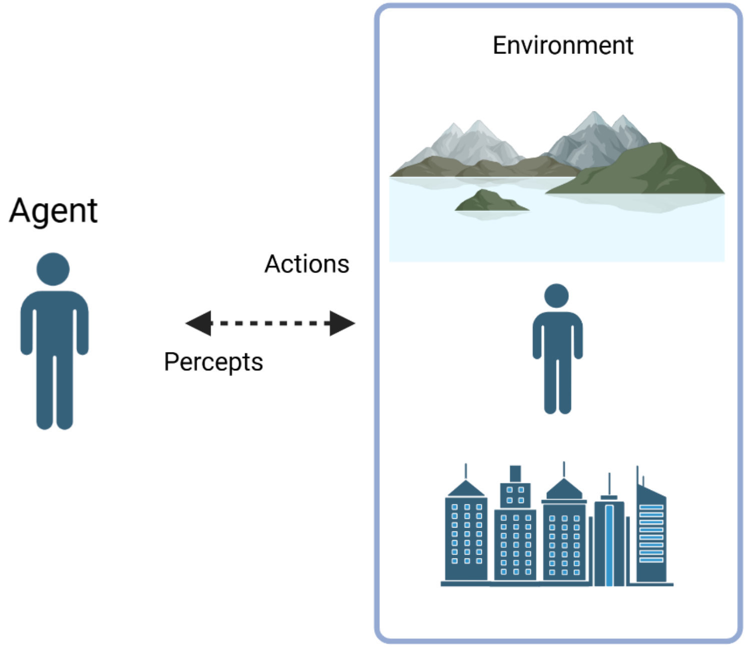

- Perception
- Action
- Decision-making
- Autonomy

Any entity capable of perceiving its environment and taking actions can be considered an agent (Cheng et al., 2024)

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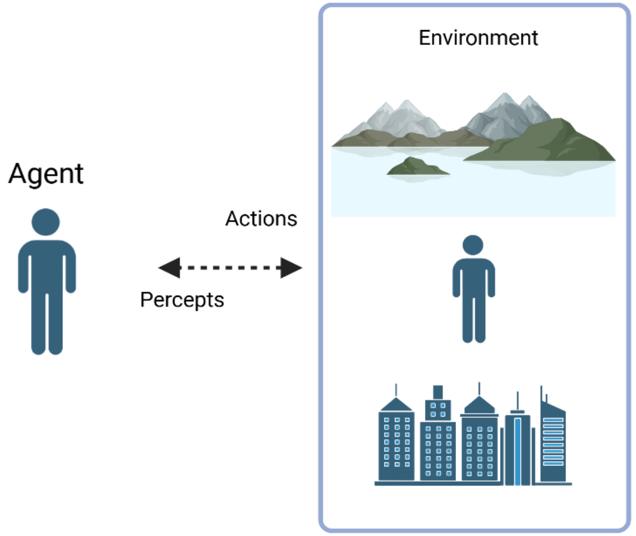

Simple Reflex Agents

Model-based Reflex Agents

Goal-based Agents

Utility-based Agents

Learning Agents

(Russel and Norvig, 2010; Cheng et al., 2024)
https://www.digitalocean.com/resources/articles/types-of-ai-agents

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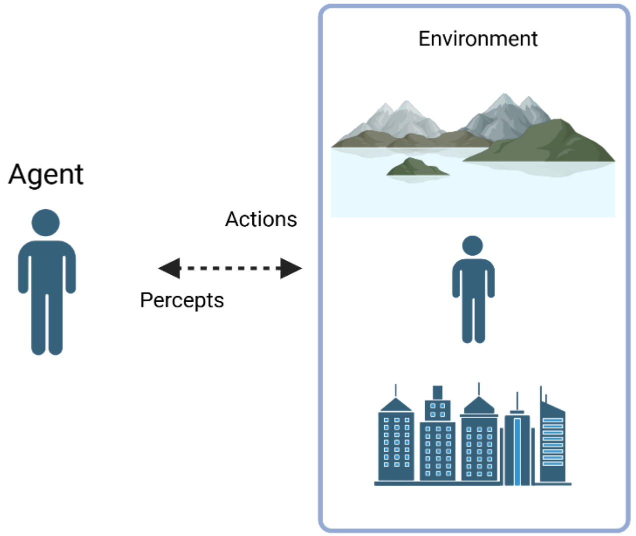

Actions based solely on current percepts

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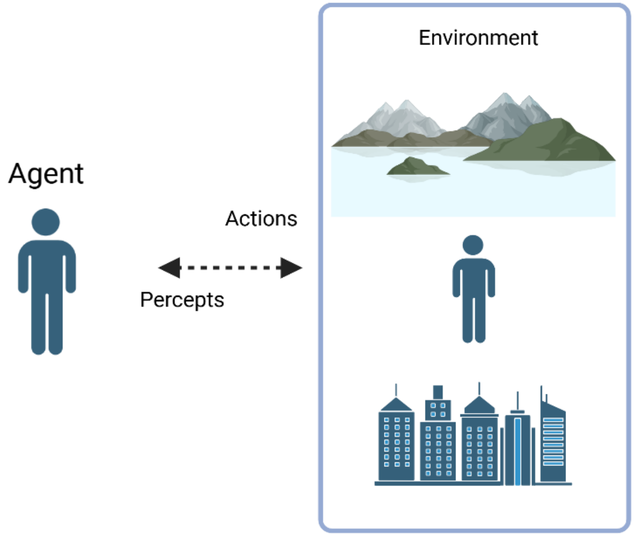

Actions based solely on current percepts

Environment model + condition-action rules

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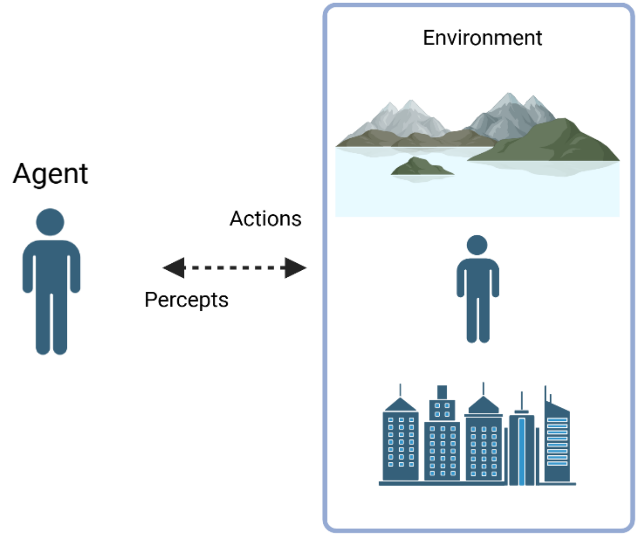

Actions based solely on current percepts

Environment model + condition-action rules

Environment model + goal-based state evaluation

Utility-based Agents

Learning Agents

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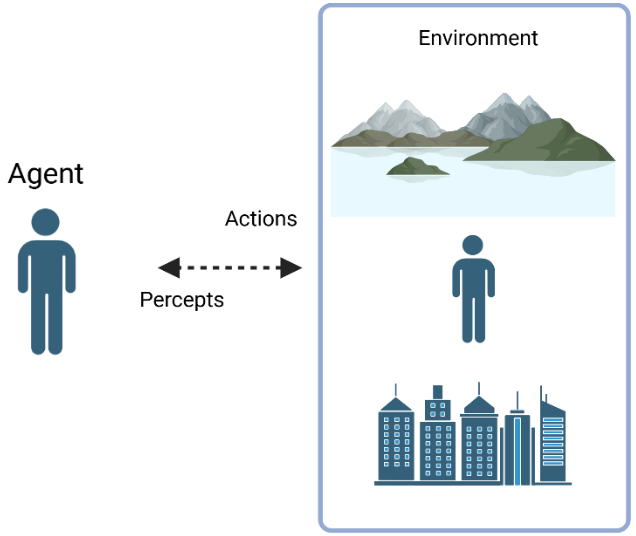

- Actions based solely on current percepts
- Environment model + condition-action rules
- Environment model + goal-based state evaluation
- Environment model + utility function for state evaluation with trade offs
- Learning Agents

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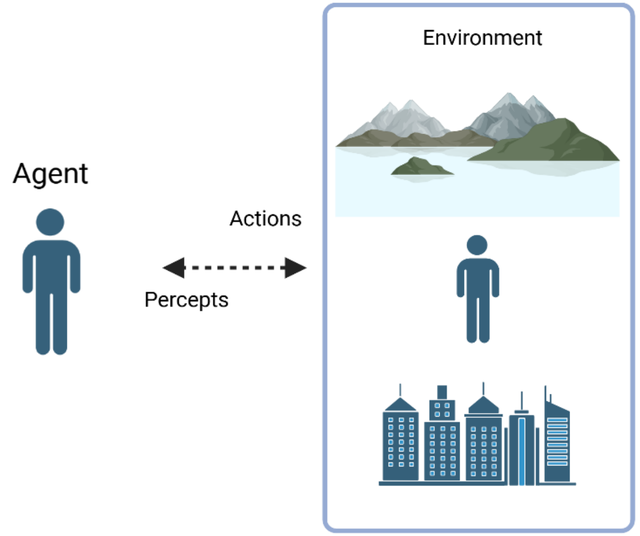

Simple Reflex Agents	Model-based Reflex Agents	Goal-based Agents
Utility-based Agents	Reinforcement Learning based agents	Large Language Model based agents

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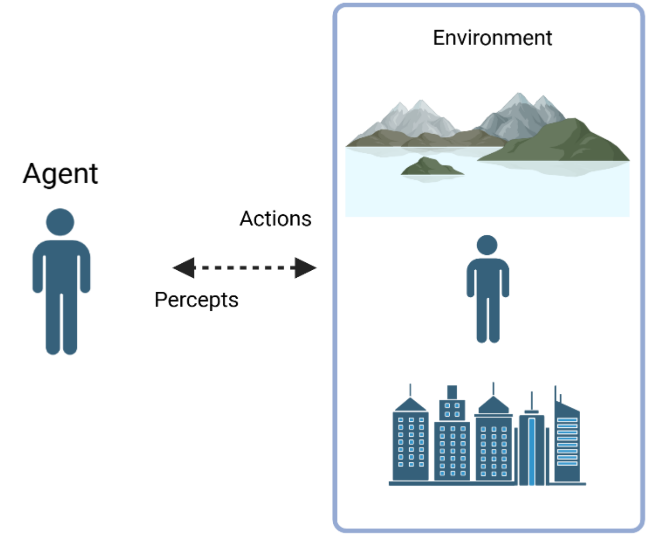

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Capacity to learn and improve based on experience

Autonomous learning capabilities and multi-step decision problems

(Russel and Norvig, 2010; Cheng et al., 2024)
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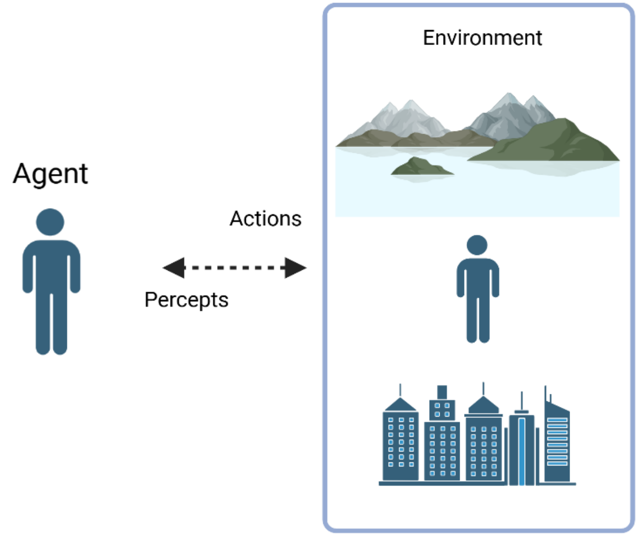
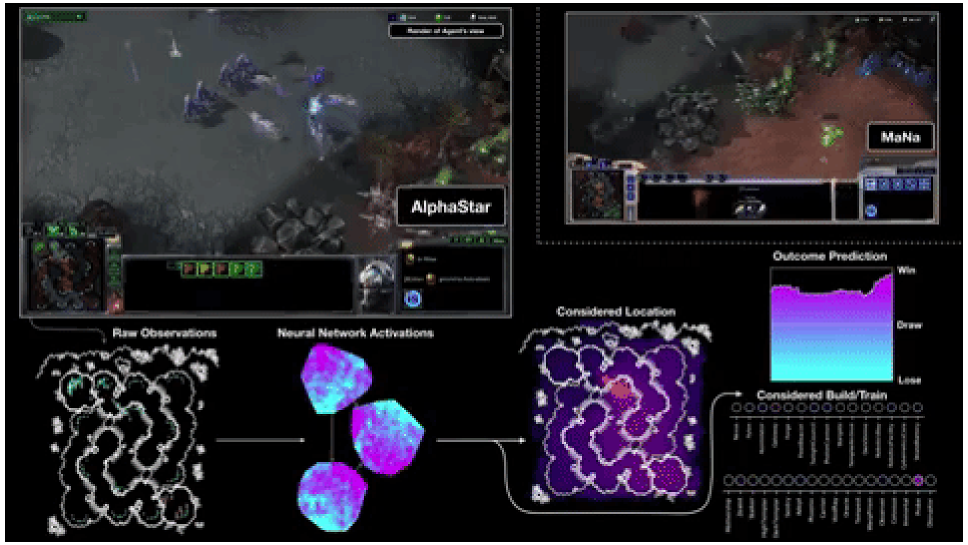


Silver et al. 2018

Learn unknown policy to maximize cumulative rewards

Teslamag, 2017



DeepMind, 2019

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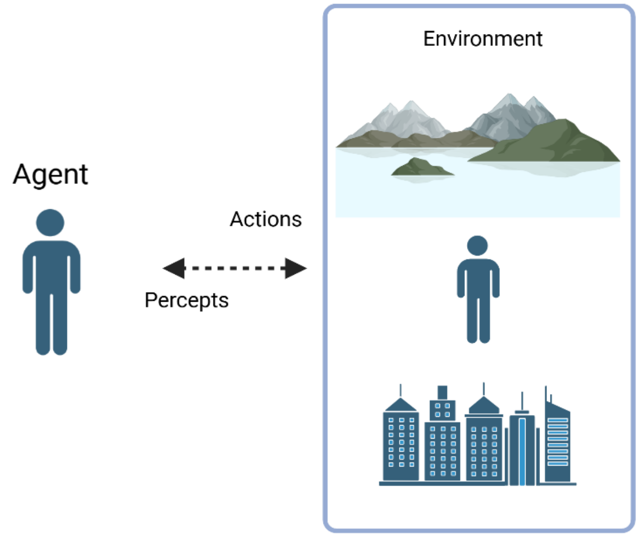

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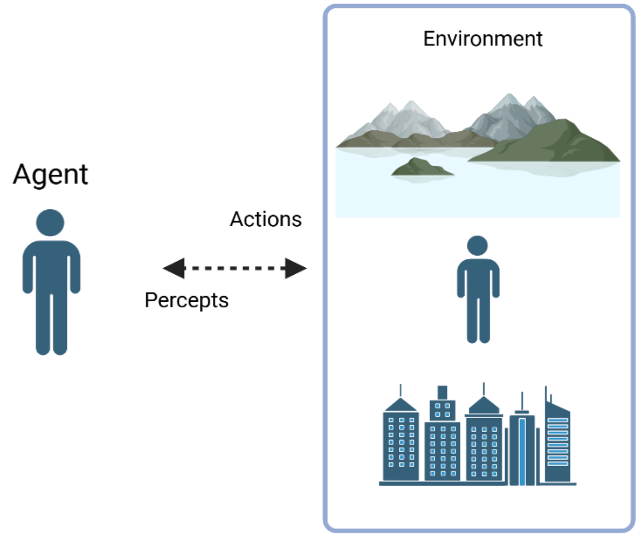

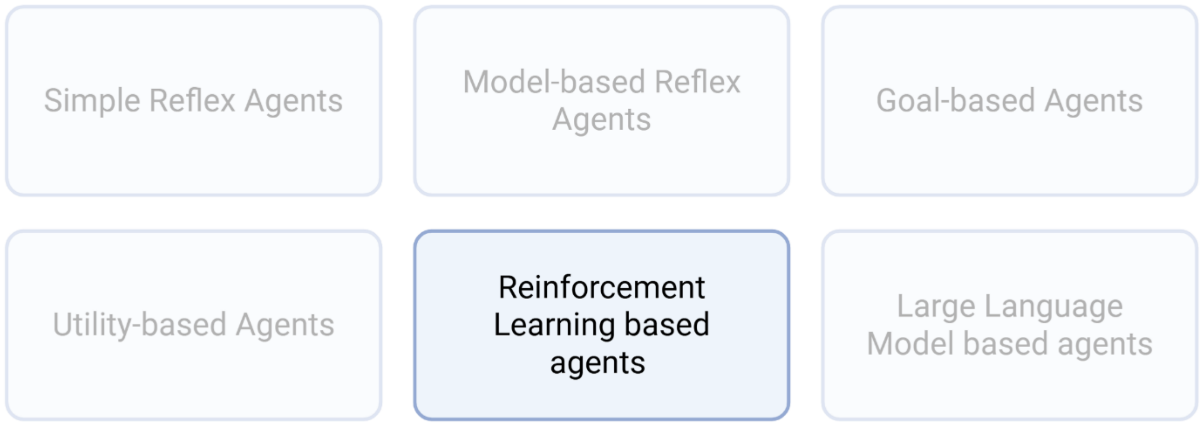

Limitations:

- Long training time
- Sample efficiency (high number of samples needed)
- Instability in non-stationary environments (change of the dynamics as policy evolves)
- Lack of generalizability (partially tackled with transfer learning, but still an active research area)

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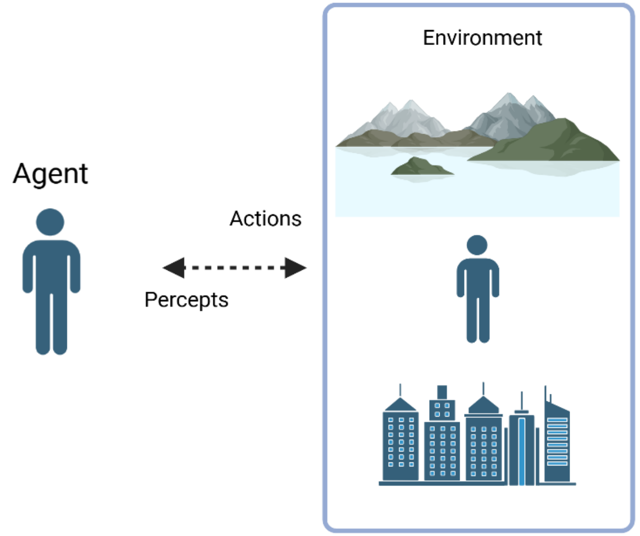

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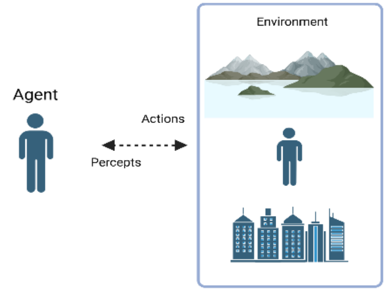

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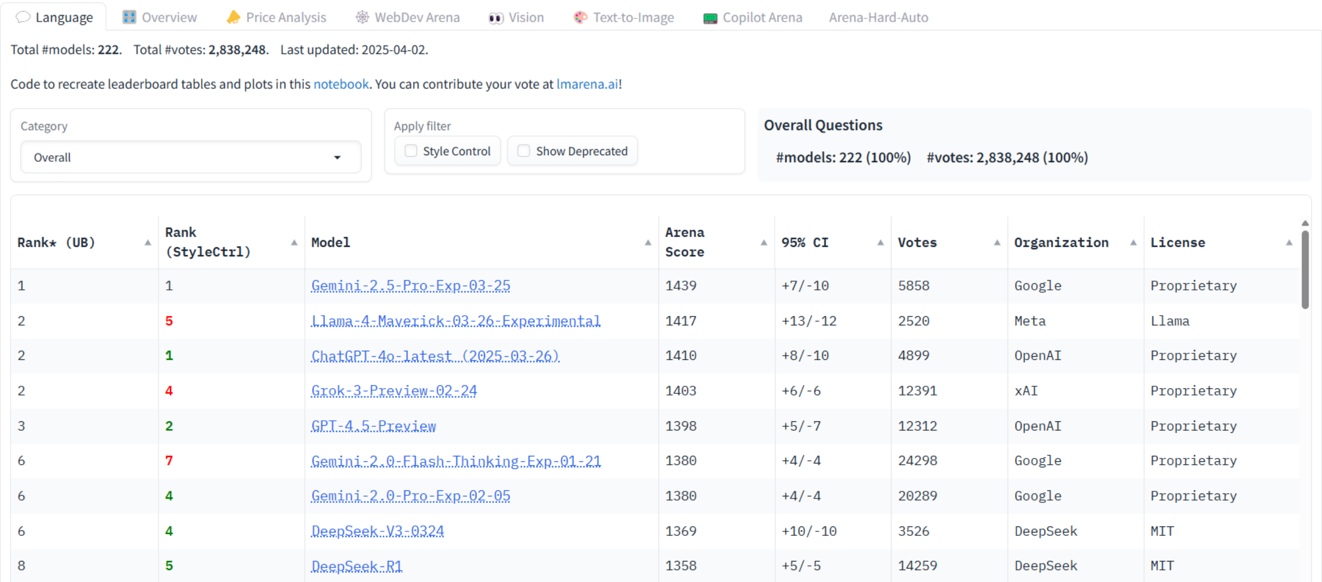
Model Name	Developer	Country of Origin
GPT-4.5	OpenAI	United States
DeepSeek R1	DeepSeek	China
Claude 3.7 Sonnet	Anthropic	United States
Grok-3	xAI	United States
Mistral Large 2	Mistral AI	France
PaLM 2	Google	United States
Gemini 2.0 Pro	Google DeepMind	United States
Llama 4 Maverick	Meta	United States
Command R+	Cohere	Canada
ERNIE 4.0	Baidu	China


Language | Overview | Price Analysis | WebDev Arena | Vision | Text-to-Image | Copilot Arena | Arena-Hard-Auto

Total #models: 222. Total #votes: 2,838,248. Last updated: 2025-04-02.

Code to recreate leaderboard tables and plots in this notebook. You can contribute your vote at lmarena.ai!

Category: Overall | Apply filter: Style Control, Show Deprecated | Overall Questions: #models: 222 (100%) #votes: 2,838,248 (100%)

Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License
1	1	Gemini-2.5-Pro-Exp-03-25	1439	+7/-10	5858	Google	Proprietary
2	5	Llama-4-Maverick-03-26-Experimental	1417	+13/-12	2520	Meta	Llama
2	1	ChatGPT-4o-latest (2025-03-26)	1410	+8/-10	4899	OpenAI	Proprietary
2	4	Grok-3-Preview-02-24	1403	+6/-6	12391	xAI	Proprietary
3	2	GPT-4.5-Preview	1398	+5/-7	12312	OpenAI	Proprietary
6	7	Gemini-2.0-Flash-Thinking-Exp-01-21	1380	+4/-4	24298	Google	Proprietary
6	4	Gemini-2.0-Pro-Exp-02-05	1380	+4/-4	20289	Google	Proprietary
6	4	DeepSeek-V3-0324	1369	+10/-10	3526	DeepSeek	MIT
8	5	DeepSeek-R1	1358	+5/-5	14259	DeepSeek	MIT

https://lmarena.ai/?leaderboard

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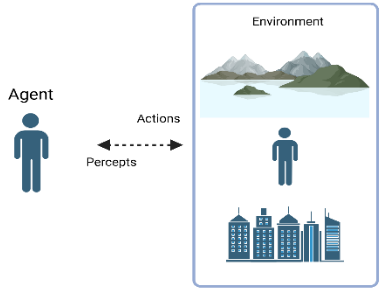

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Model	Organization	Global Average	Reasoning Average	Coding Average	Mathematics Average	Data Analysis Average	Language Average
gemini-2.5-pro-exp-03-25	Google	82.71	89.75	85.87	90.20	79.89	67.82
deepseek-r1	DeepSeek	69.79	83.17	66.74	80.71	69.78	48.53
o1-2024-12-17-high	OpenAI	74.49	91.58	69.69	80.32	65.47	65.39
claude-3-7-sonnet-thinking	Anthropic	75.07	87.83	74.54	79.00	74.05	59.93
qwq-32b	Alibaba	69.99	83.50	72.23	77.82	65.03	51.35
o3-mini-2025-01-31-high	OpenAI	74.19	89.58	82.74	77.29	70.64	50.68
gemini-2.0-flash-thinking-exp-01-21	Google	63.81	78.17	53.49	75.85	69.37	42.18
deepseek-v3-0324	DeepSeek	63.94	65.83	70.90	73.52	60.37	49.07
o3-mini-2025-01-31-medium	OpenAI	67.38	86.33	65.38	72.37	66.56	46.26
gemini-exp-1206	Google	61.44	57.00	63.41	72.36	63.16	51.29
gemini-2.0-pro-exp-02-05	Google	61.48	60.08	63.49	70.97	68.02	44.85

https://livebench.ai/#/

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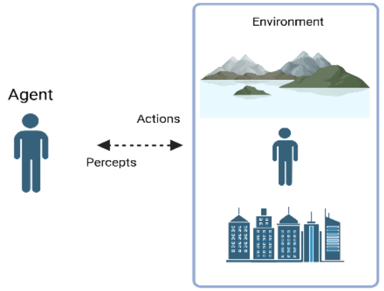

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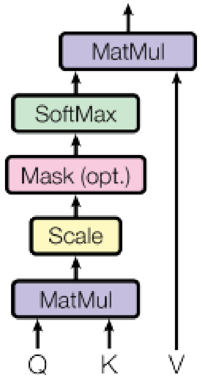
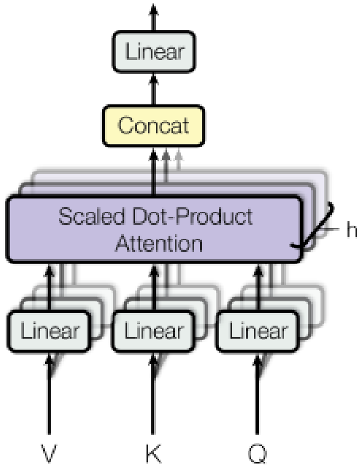
Attention is all you need (Vaswani et al., 2017)

Scaled Dot-Product Attention



Multi-Head Attention



- Text => Embedding
- **Attention block** captures the context:

Embedding of Part 1 ⇔Embedding of Part 2 etc.

- **Feed forward block** links the data with

the stored facts: Embedding ⇔ facts

- Embedding => Text

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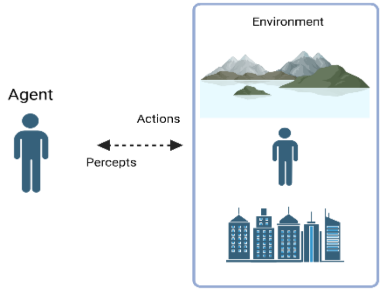

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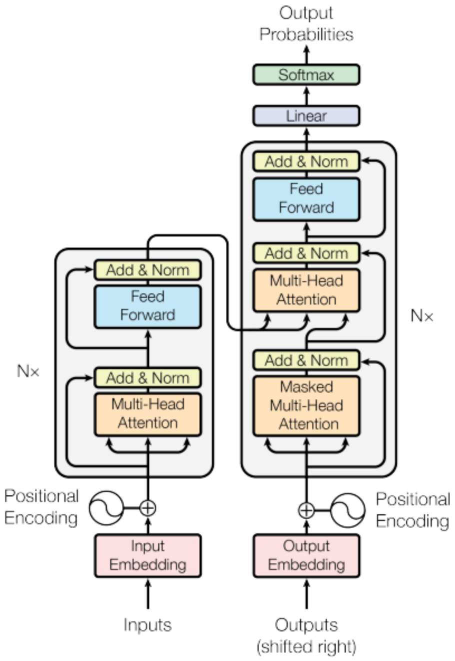

Figure 1: The Transformer - model architecture.

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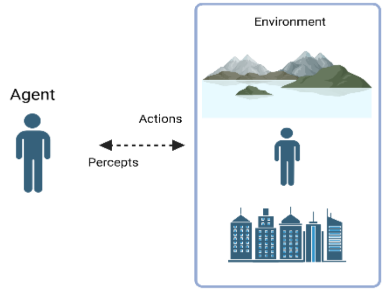

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$$\frac{e^{x_i}/T}{\sum_{n=0}^{N-1} e^{x_n}/T}$$

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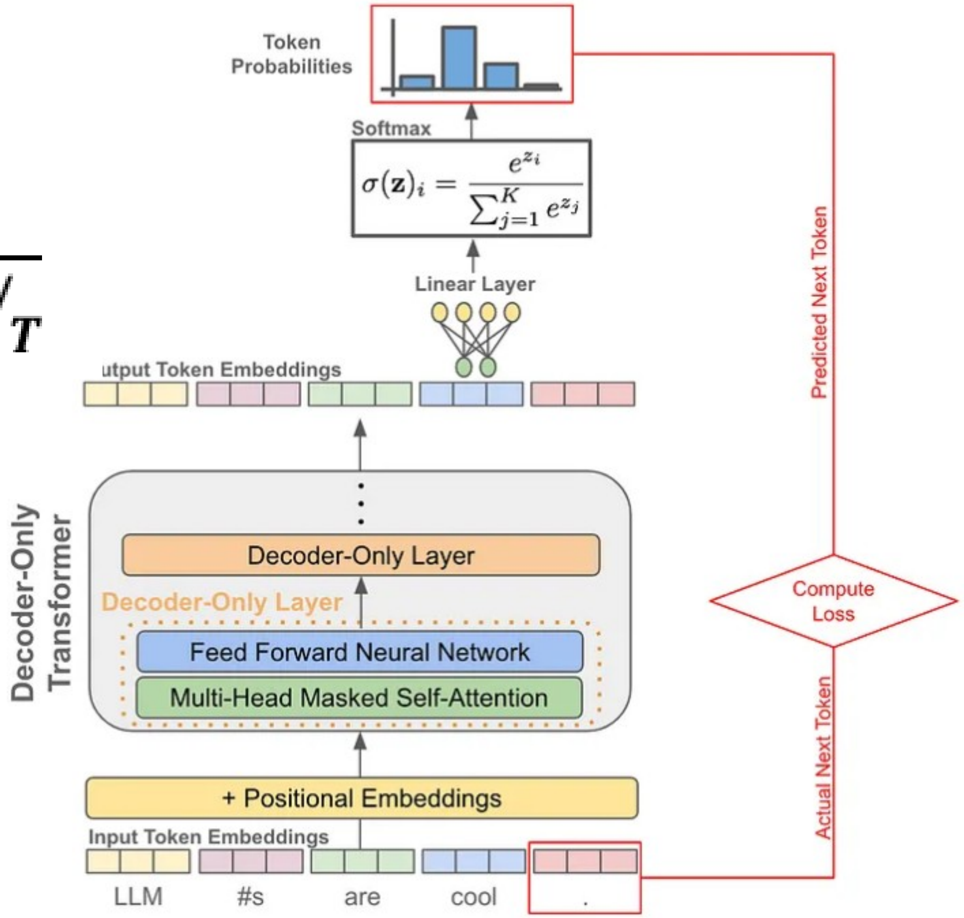

Figure from Hongying et al.2025

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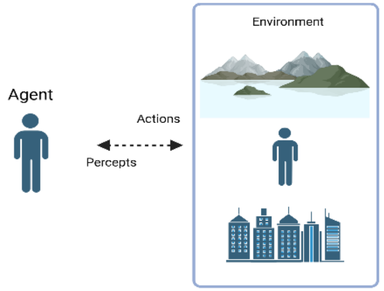

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N-shots prompting (N refers to the example numbers)

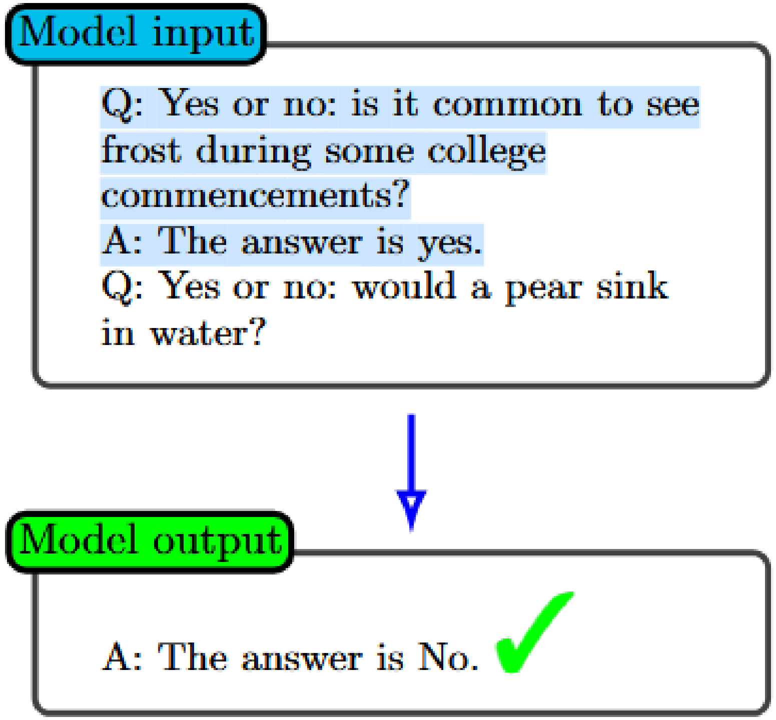

Figure from Chen et al. 2023

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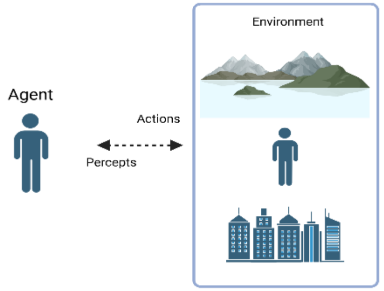

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Chain of thoughts

Model input

Imagine an infinitely wide entrance, which is more likely to pass through it, a military tank or a car? Let's think step by step.

Model output

Given an infinitely wide entrance, both a military tank and a car could pass through without any issues related to the width. However, there are still several factors that could influence which is more likely to pass through:

1. Ground surface: tanks are designed to handle rough terrains that cars may struggle with. If the ground leading to and from the entrance is too rugged or steep for a car, the tank would be more likely to pass through.
2. Weight restrictions: if the entrance has a load limit due to the materials used in its construction, the car, being lighter, could be more likely to pass through. The average car weighs around 1.5 tons, while a military tank can weigh anywhere from 60 to 70 tons.
3. Height clearance: if the entrance is infinitely wide but not infinitely tall, the height of the vehicle matters. The average height of a car is around 1.5 meters, while tanks can be around 2.4 meters. If the entrance is shorter than around 2.4 meters, a car would be more likely to pass through.
4. Regulations and permissions:

...

In conclusion, without specific information about the conditions at the entrance, we would assume that a car is more likely to pass through, simply because cars are more common, more likely to meet potential height and weight restrictions and face fewer regulatory barriers.

Figure from Chen et al. 2023

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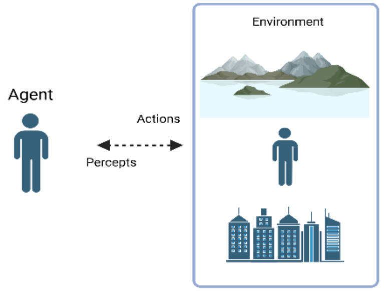

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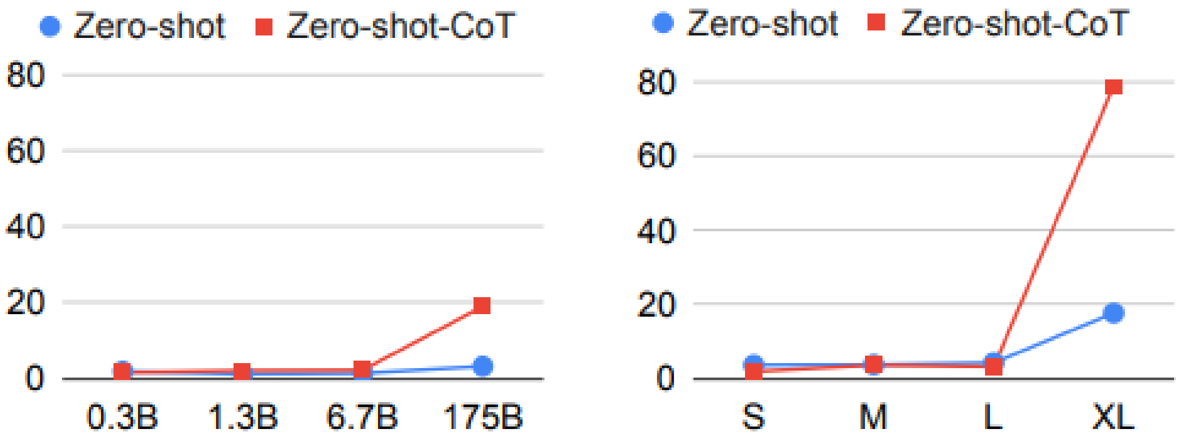
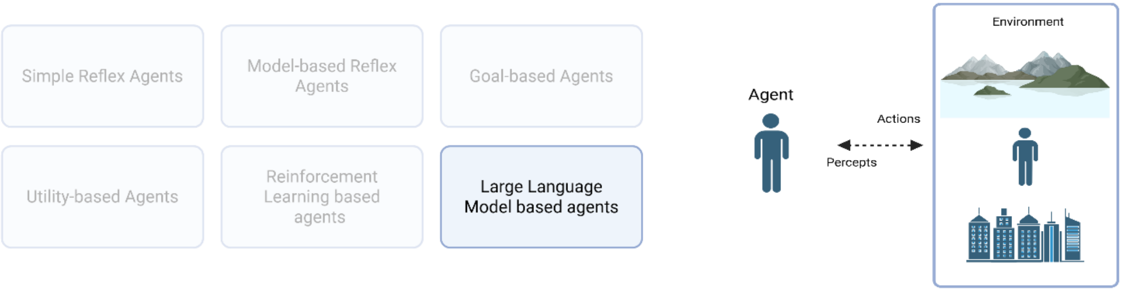


(a) MultiArith on Original GPT-3 (b) MultiArith on Instruct GPT-3

Figures are from Kojima et al., 2022

Part I. LLM-based agents



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Limitations:

- Context Length Constraint

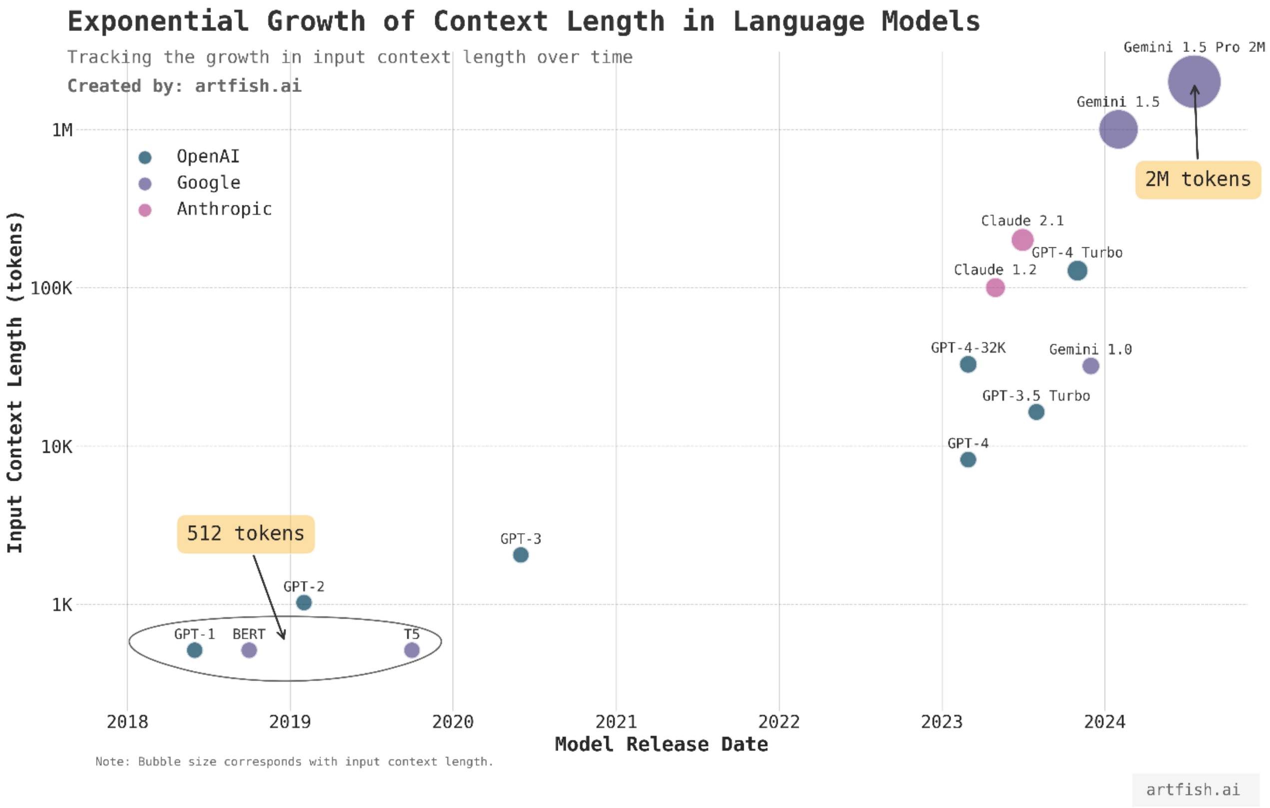

Figure from https://www.artfish.ai/p/long-context-llms

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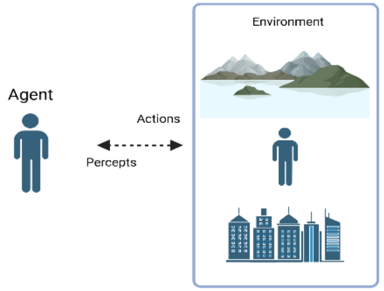

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Lost in the Middle: How Language Models Use Long Context

Nelson F. Liu[1*] **Kevin Lin**[2] **John Hewitt**[1] **Ashwin Paranjape**[3] **Michele Bevilacqua**[3] **Fabio Petroni**[3] **Percy Liang**[1]

[1]Stanford University [2]University of California, Berkeley [3]Samaya AI

nfliu@cs.stanford.edu

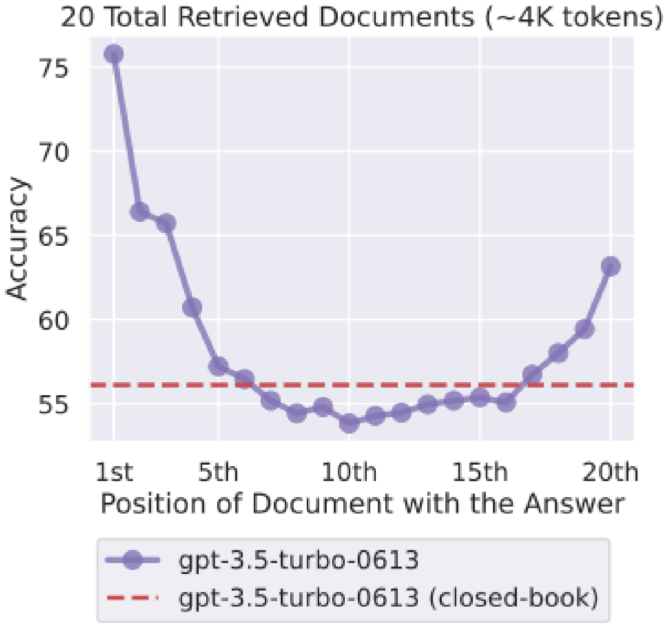
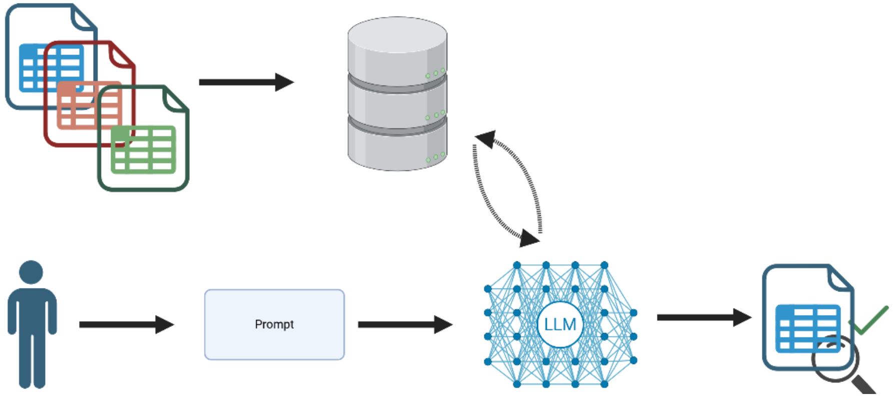
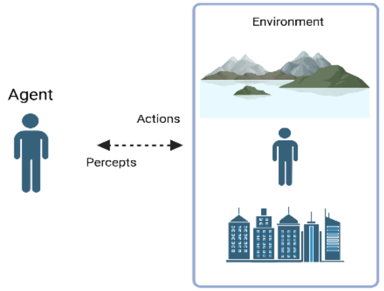


Figure from Liu et al., 2023



Images were created with biorender.com

Part I. LLM-based agents



We define AI as the study of agents that receive percepts from the environment and perform actions.

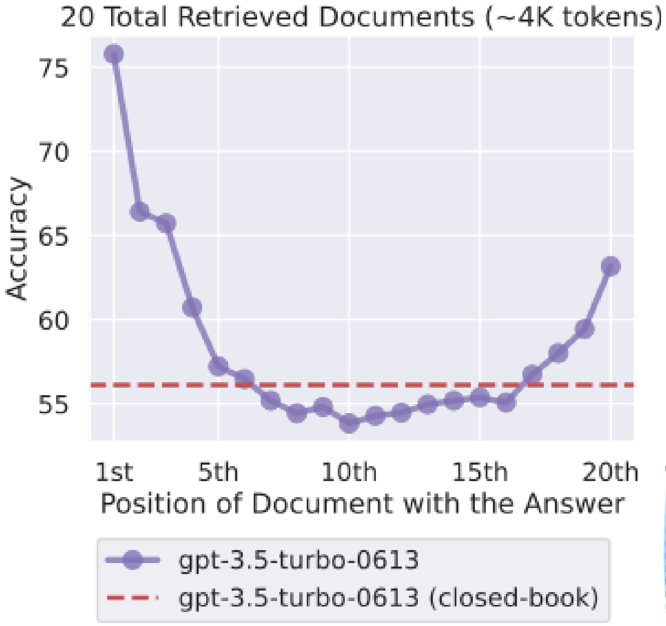
— *Artificial Intelligence: A Modern Approach*, Stuart Russell and Peter Norvig (2003).

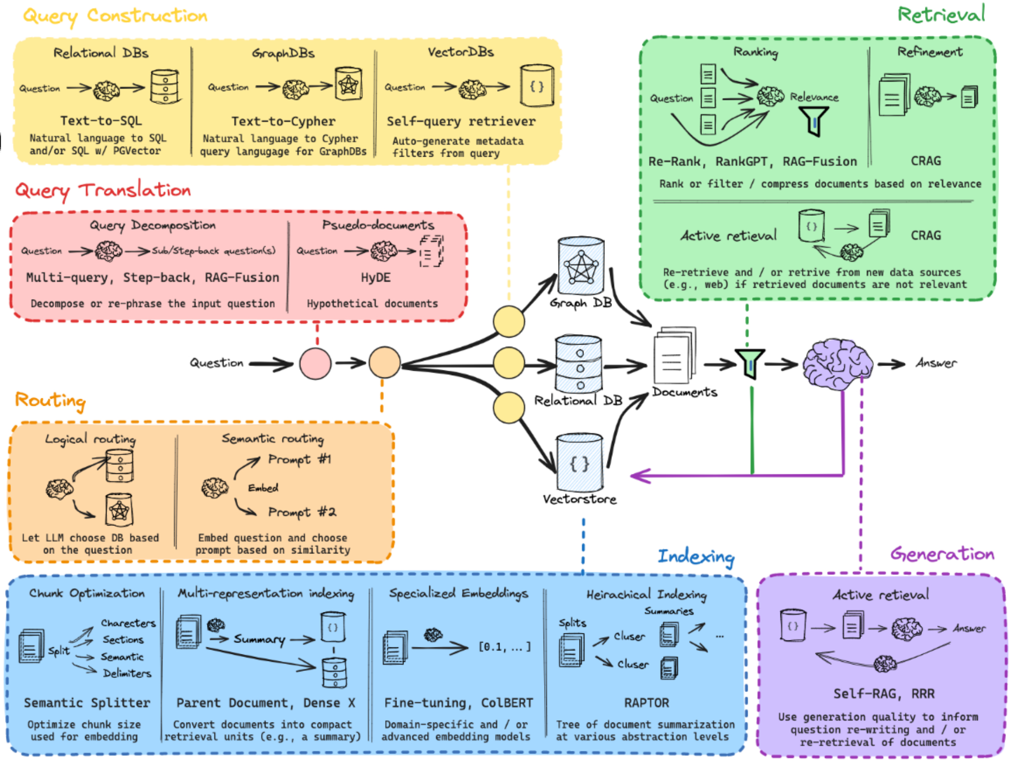
Limitations:

- Context Length Constraint

Lost in the Middle: How Language Models Use Long Contexts

Nelson F. Liu[1*] Kevin Lin[2] John Hewitt[1] Ashwin Paranjape[3] Michele Bevilacqua[3] Fabio Petroni[3] Percy Liang[1]
[1]Stanford University [2]University of California, Berkeley [3]Samaya AI
nfliu@cs.stanford.edu





RAG overview figure from
https://python.langchain.com/v0.2/docs/concepts/

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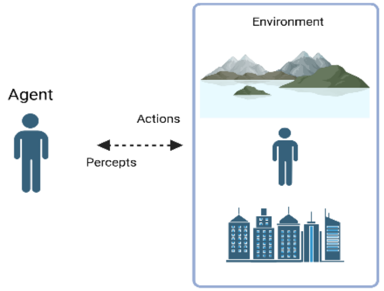
Part I. LLM-based agents



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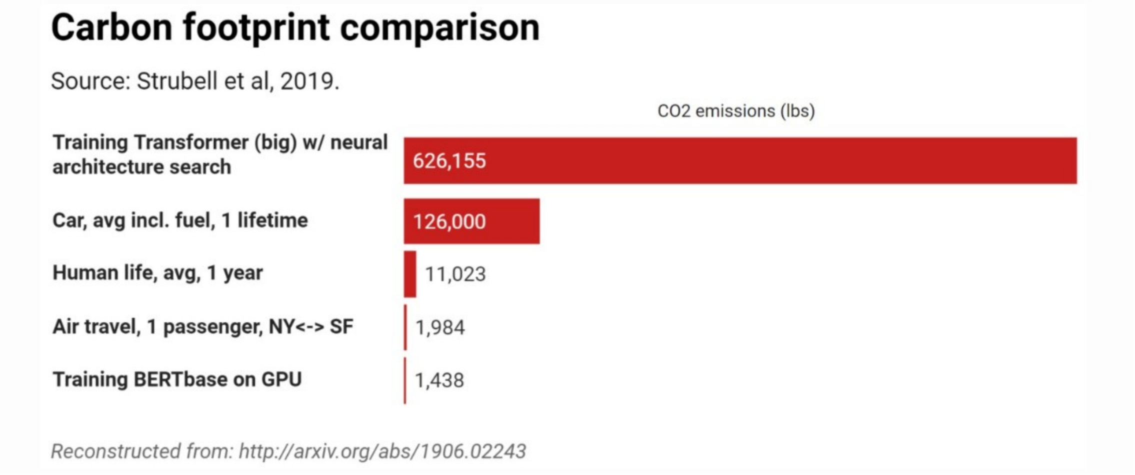
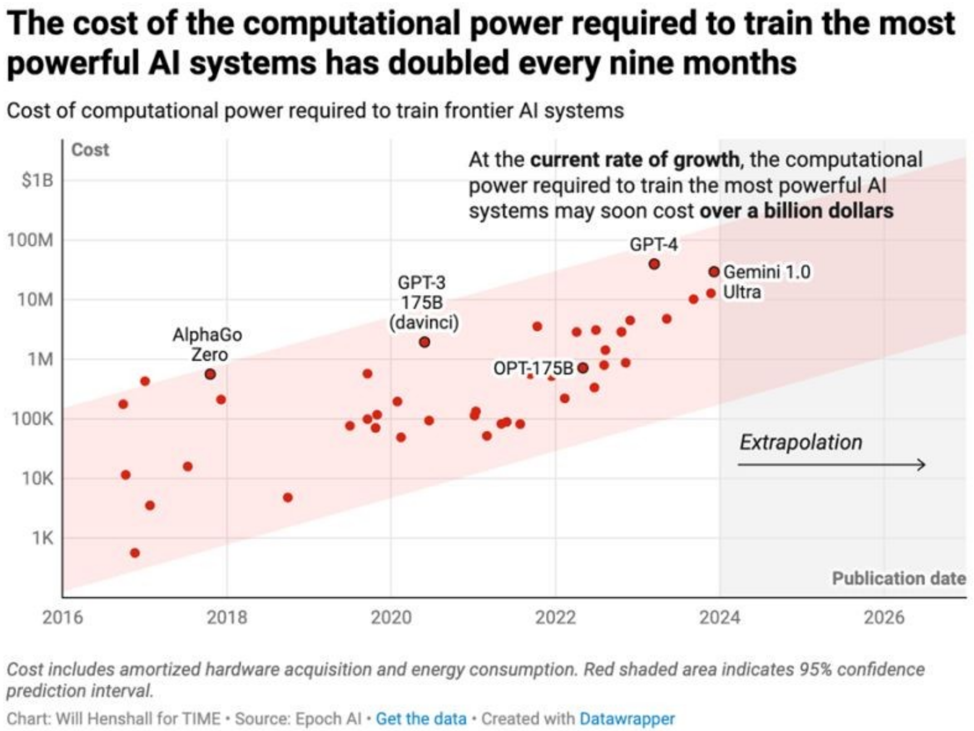
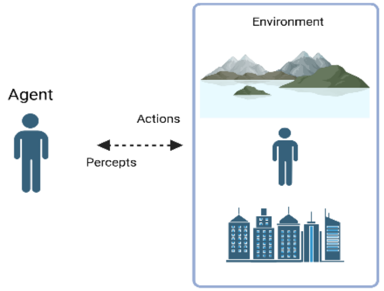
Limitations:

- Context Length Constraint
- Protracted Knowledge Update





Part I. LLM-based agents



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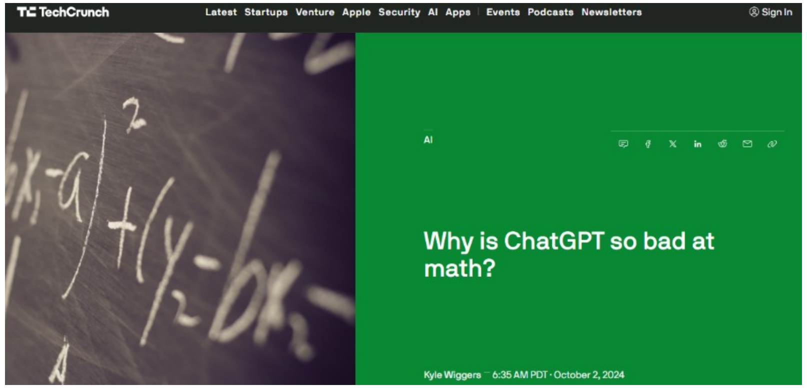
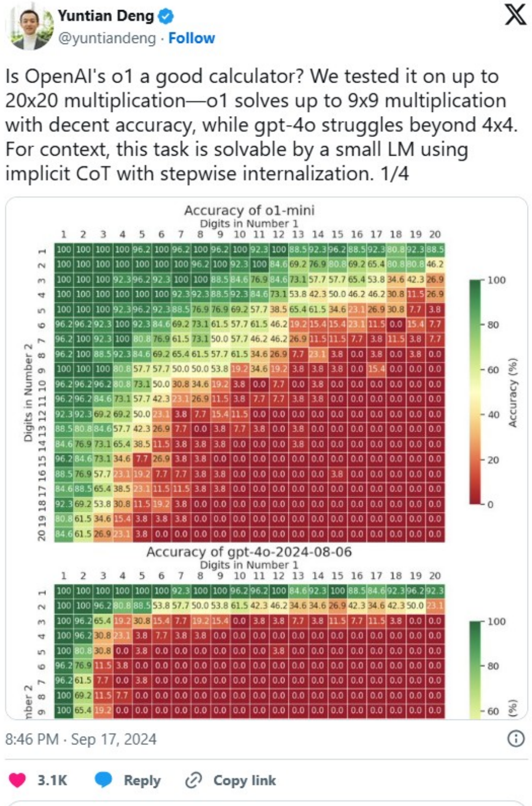
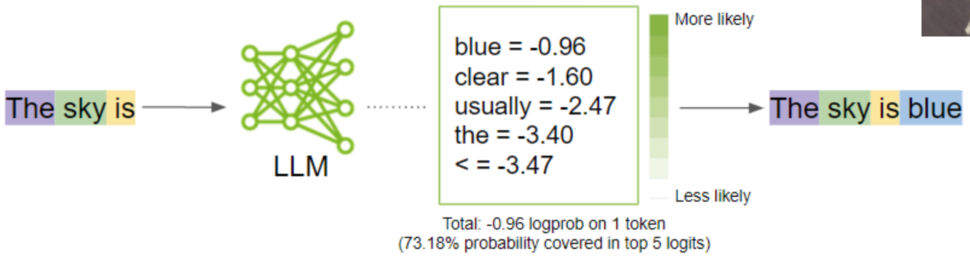
Limitations:

- Context Length Constraint
- Protracted Knowledge Update
- No direct Tool Utilization (math)

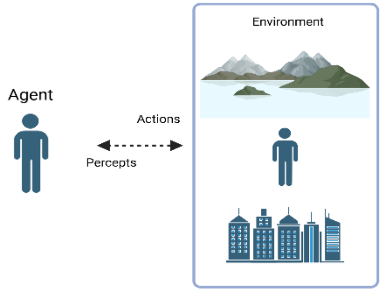






LLM overview figure from https://developer.nvidia.com/blog/how-to-get-better-outputs-from-your-large-language-model/

Part I. LLM-based agents



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Limitations:

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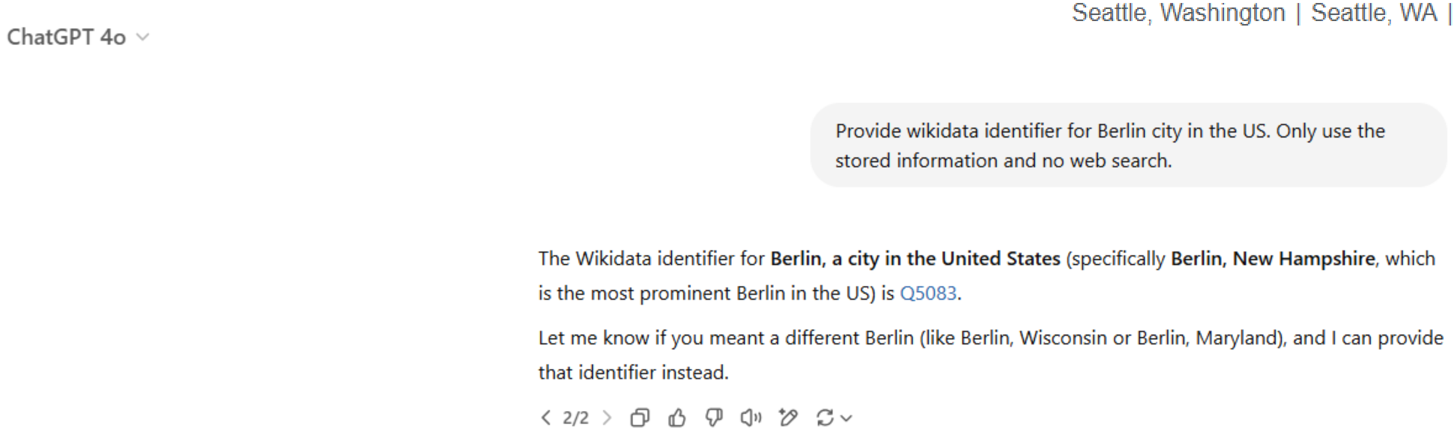

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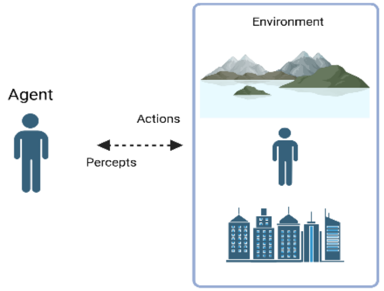

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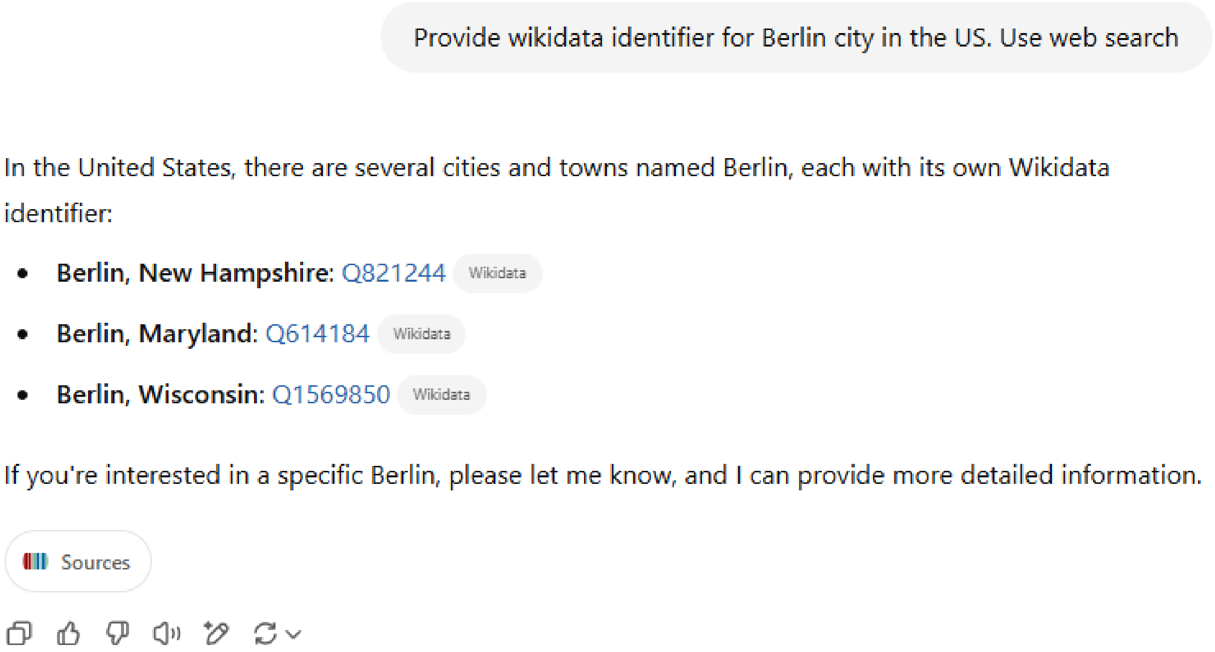

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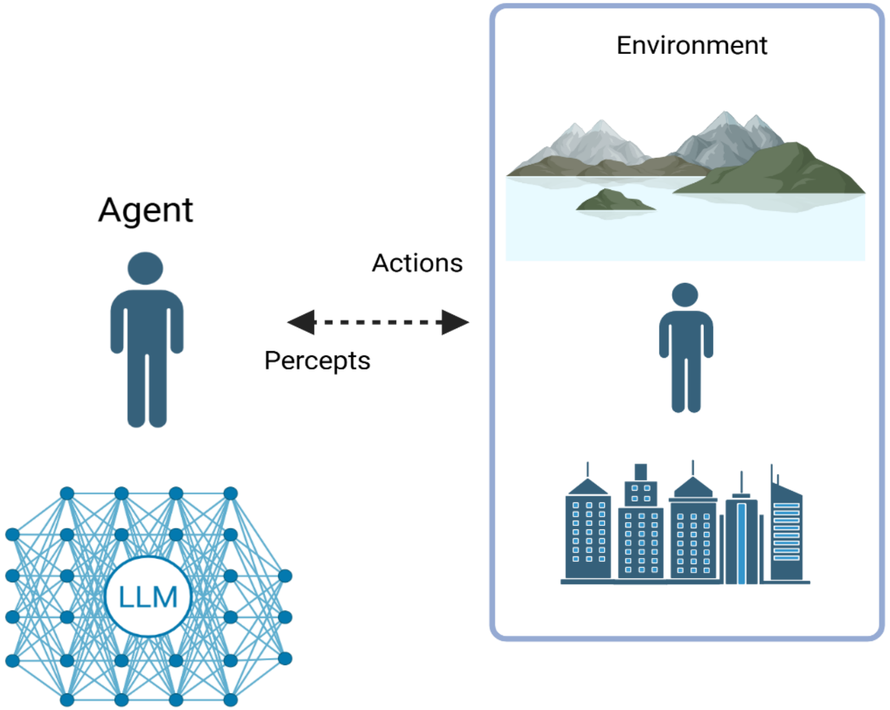

Images were created with biorender.com

Part I. LLM-based agents

Simple Reflex Agents	Model-based Reflex Agents	Goal-based Agents
Utility-based Agents	Reinforcement Learning based agents	**Large Language Model based agents**

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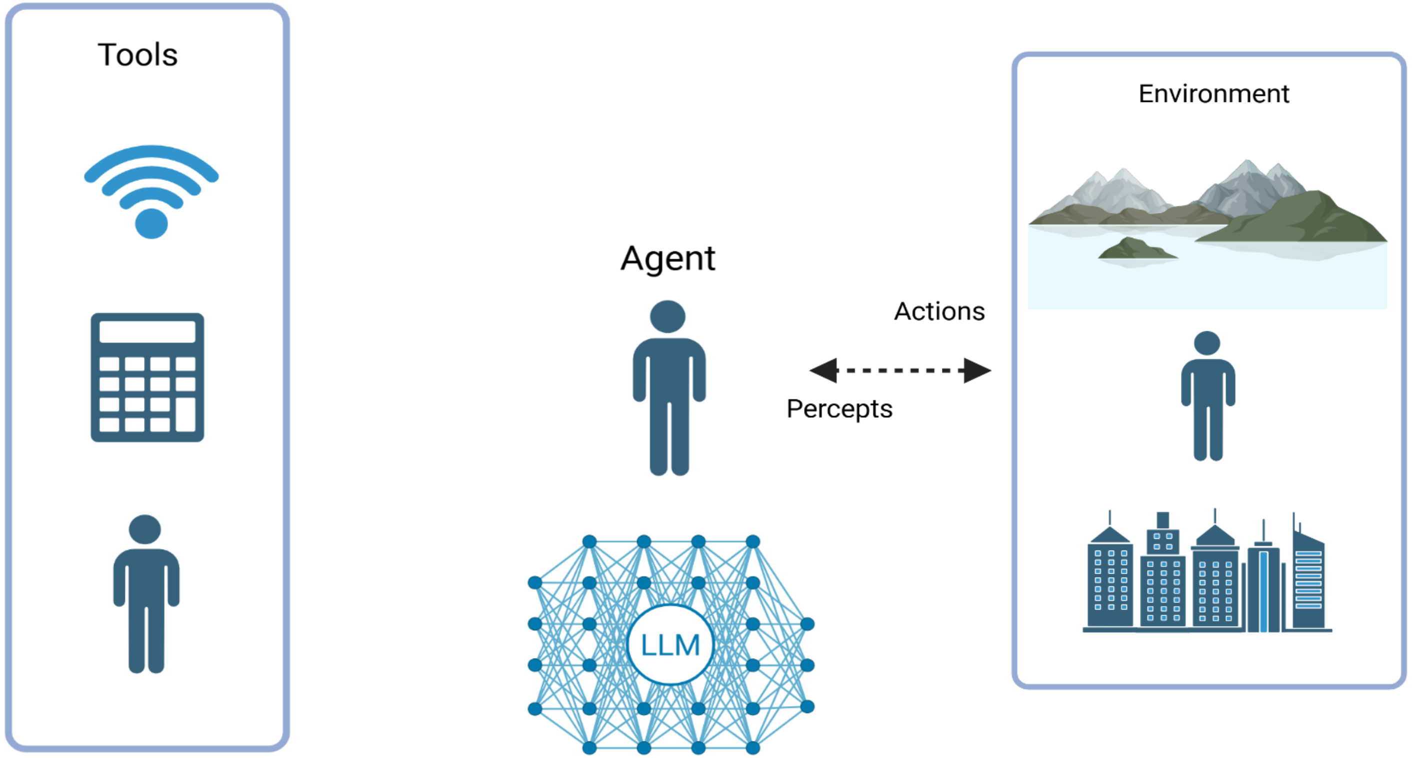

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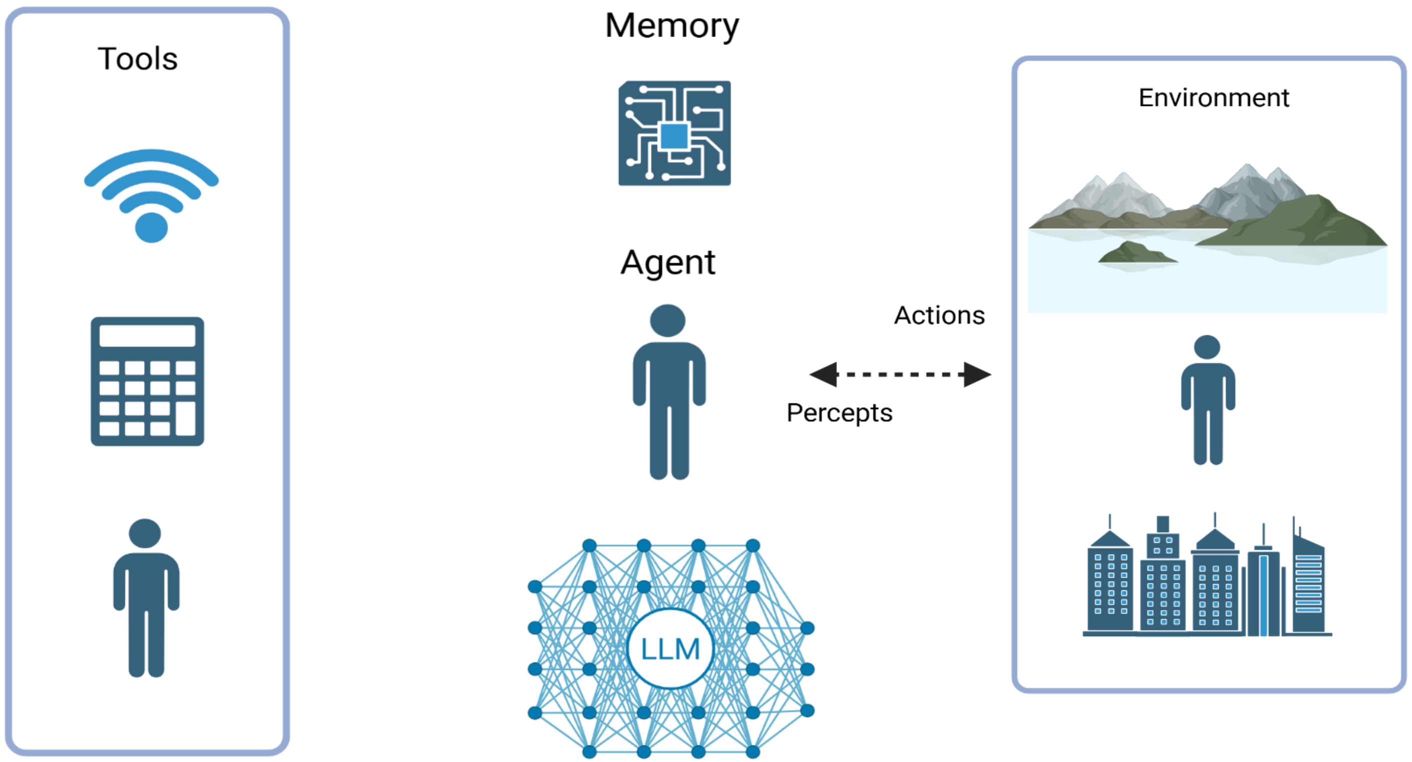

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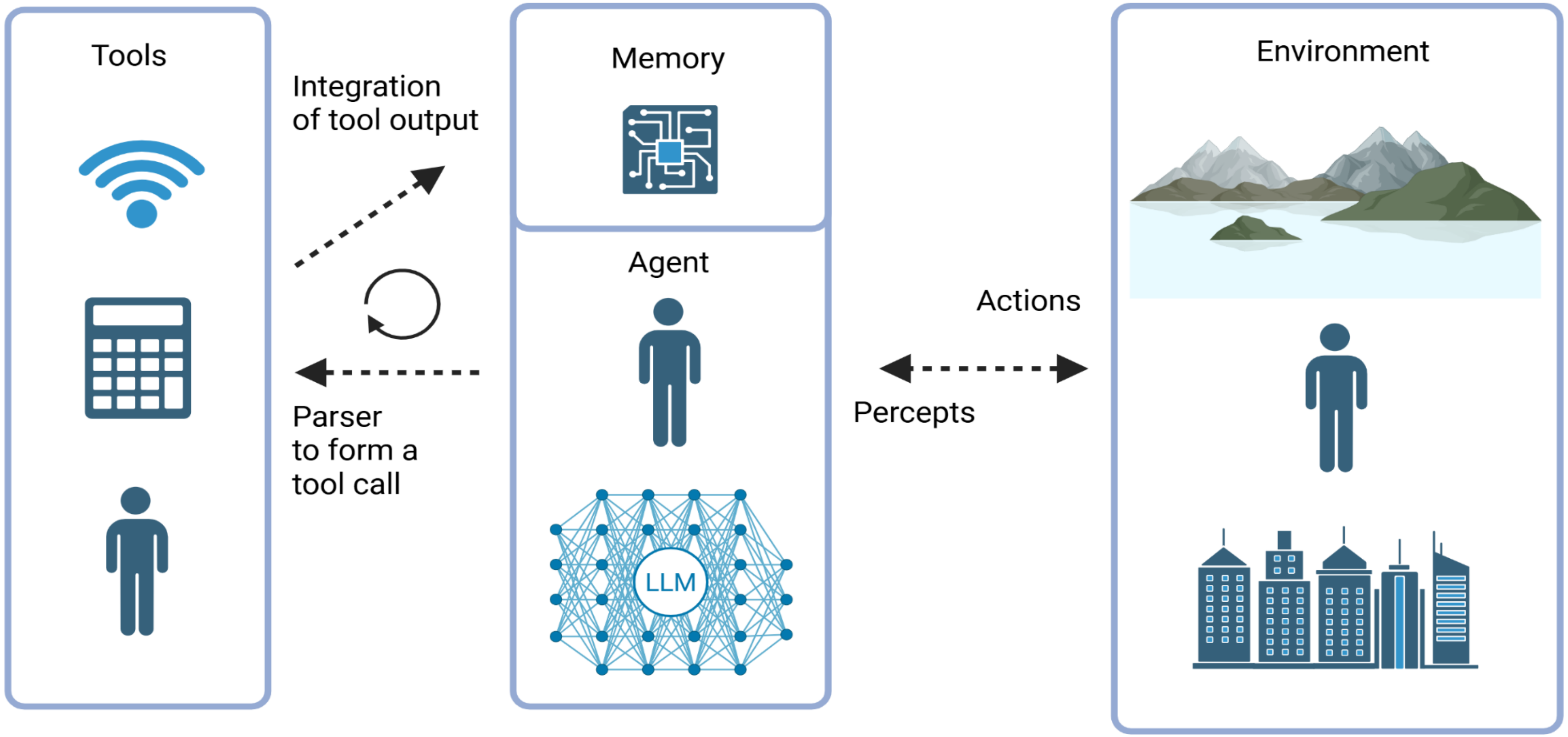

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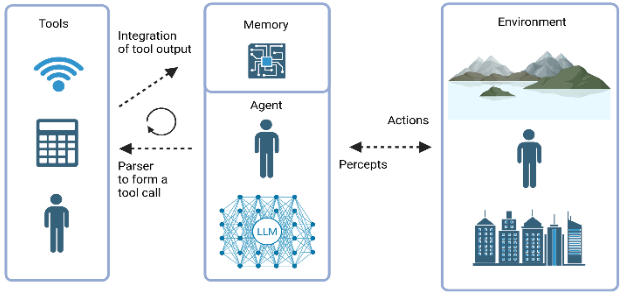

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Figures are adapted from
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Waterloo News. https://uwaterloo.ca/news/media/qa-experts-why-chatgpt-struggles-math

Agentic schemas were created with Biorender.com

Extra sources for prompt engineering

Tips and Explanations from OpenAI

https://help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-the-openai-api

Examples

https://platform.openai.com/examples

Prompt engineering tips from HuggingFace
https://huggingface.co/docs/transformers/en/tasks/prompting

Prompt engineering tips from Google

https://ai.google.dev/gemini-api/docs/prompting-intro

German Prompt Generator

https://deinkikompass.de/prompt-generator

Outline

Part I. What are AI-based autonomous agents?

Break I

Part II. Why and in which areas to use agents?

Break II

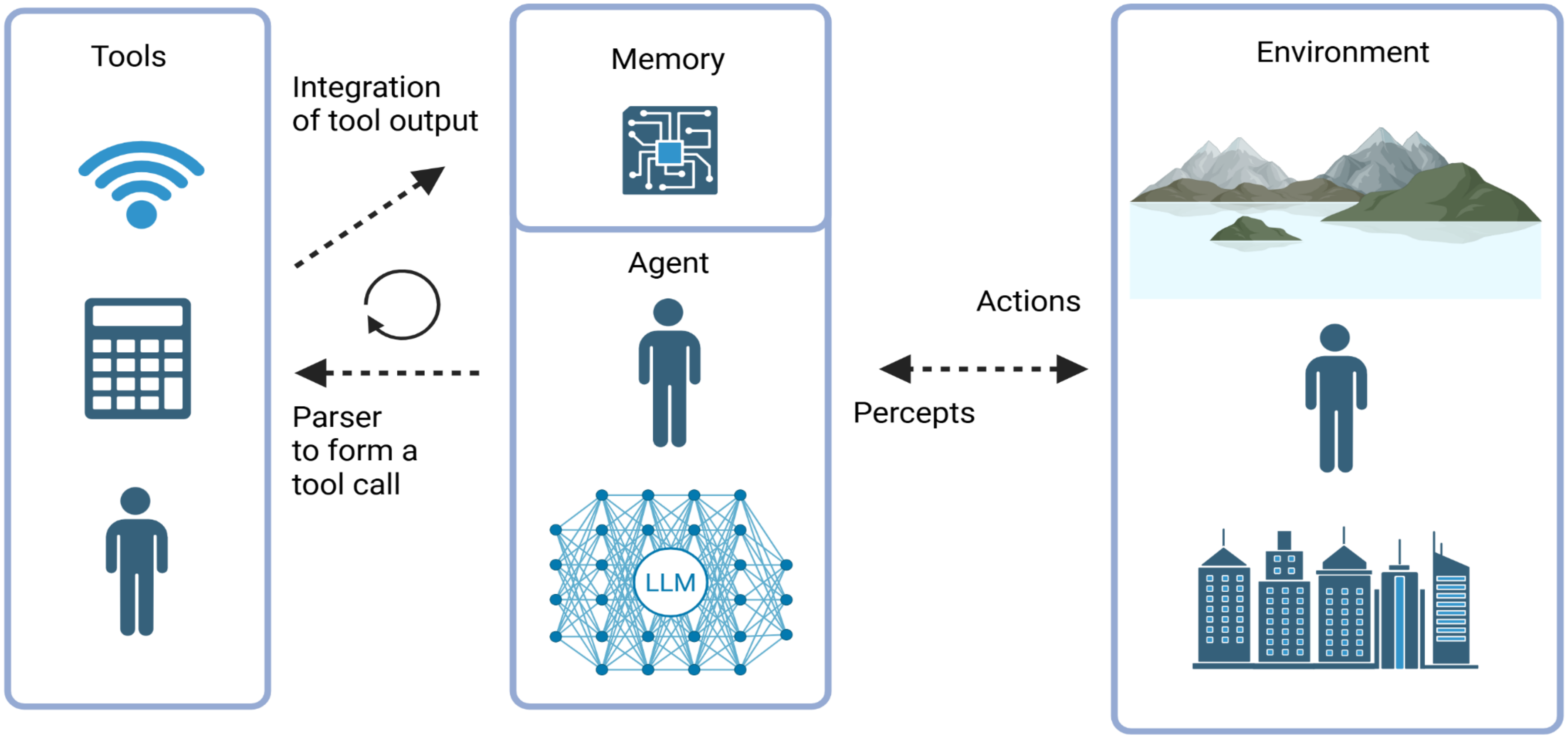
Part III. How to practically start using agents?

Outline

BfR

Part 2. Single agents

Simple Reflex Agents	Model-based Reflex Agents	Goal-based Agents
Utility-based Agents	Reinforcement Learning based agents	**Large Language Model based agents**



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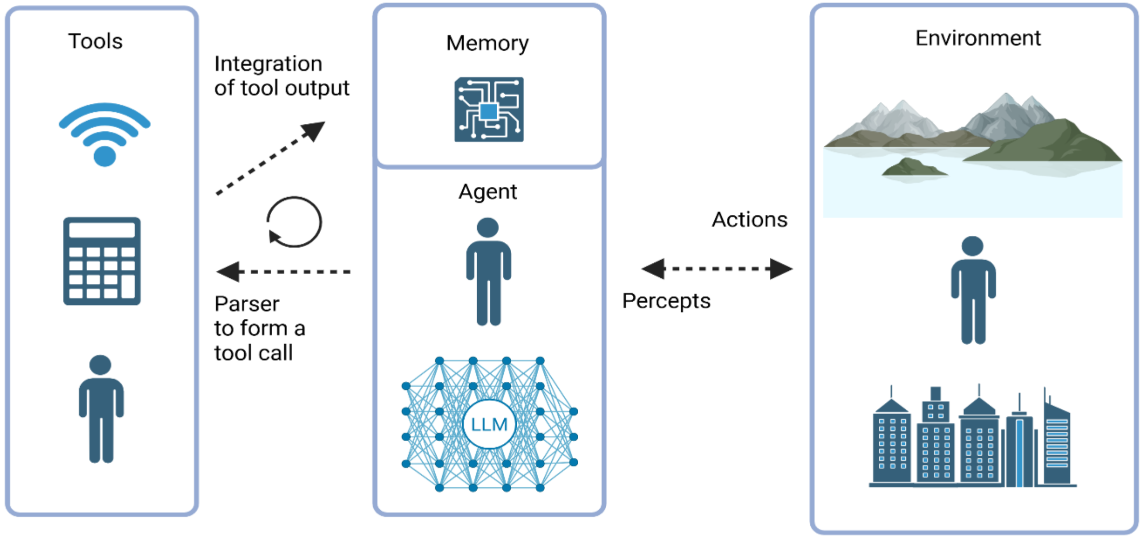
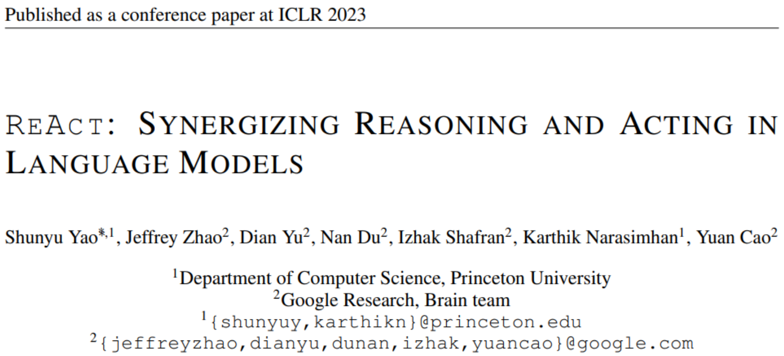
Part 2. Single Agents





Published as a conference paper at ICLR 2023

REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS

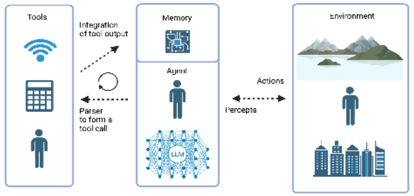
Shunyu Yao[*,1], Jeffrey Zhao[2], Dian Yu[2], Nan Du[2], Izhak Shafran[2], Karthik Narasimhan[1], Yuan Cao[2]

[1]Department of Computer Science, Princeton University
[2]Google Research, Brain team
[1]{shunyuy,karthikn}@princeton.edu
[2]{jeffreyzhao,dianyu,dunan,izhak,yuancao}@google.com

Figure from https://www.foundingminds.com/chain-of-thought-reasoning-the-magic-behind-the-o1-model/

Images were created with biorender.com

Part 2. Single Agents



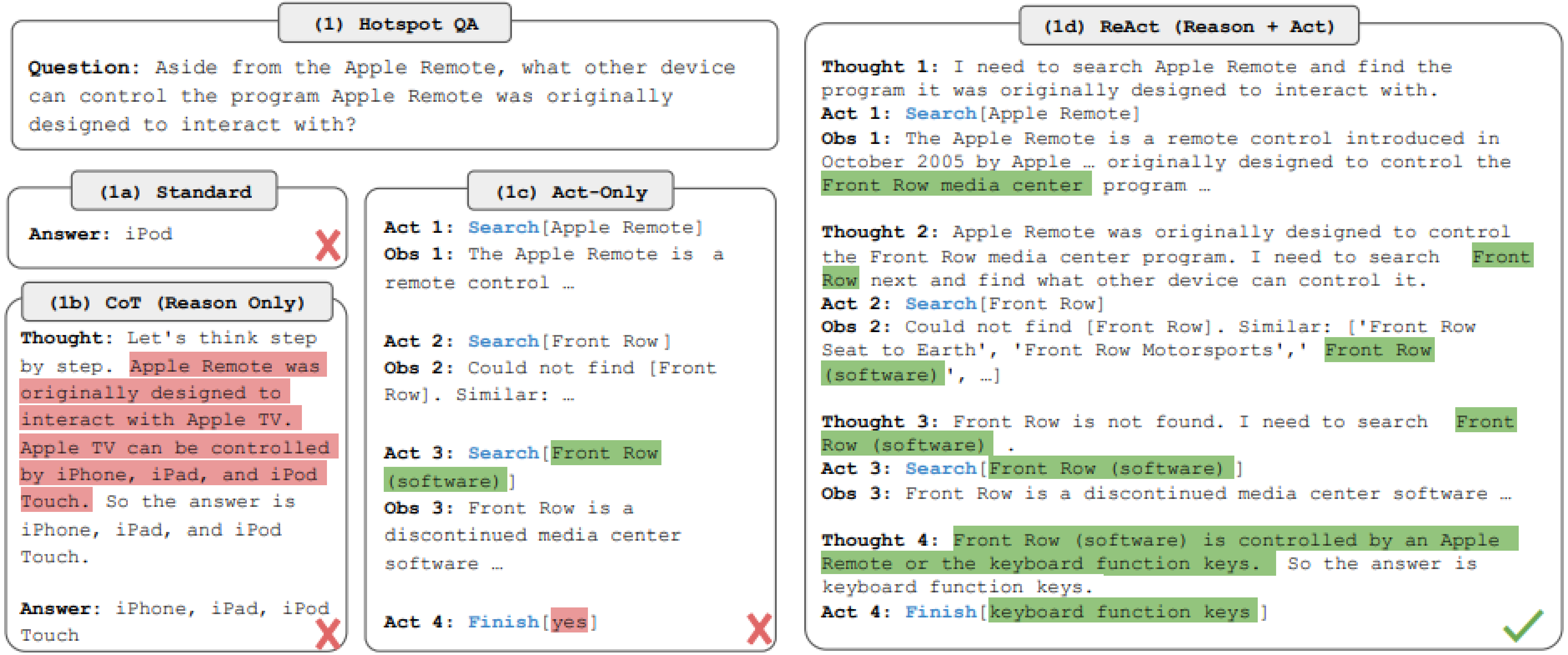


Figure from Yao et al., 2023

Part 2. Single agents

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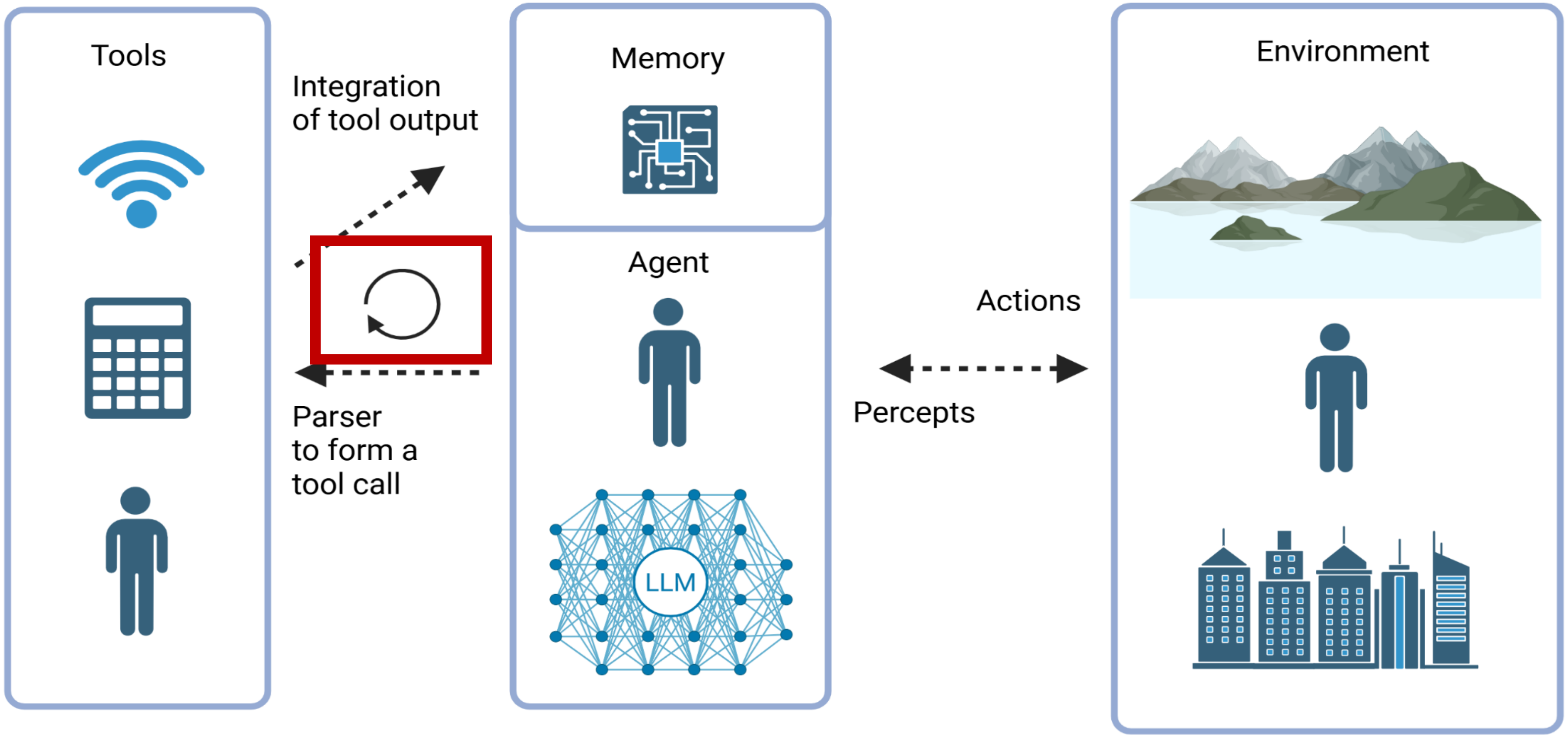

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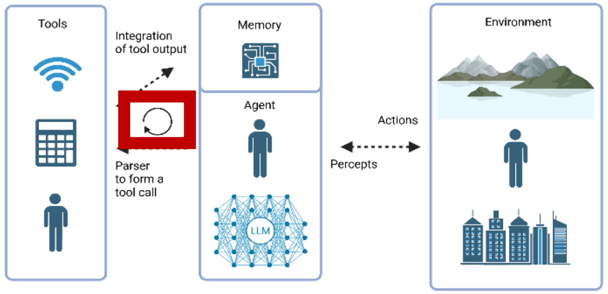

Published as a conference paper at ICLR 2024

CRITIC: LARGE LANGUAGE MODELS CAN SELF-CORRECT WITH TOOL-INTERACTIVE CRITIQUING

Zhibin Gou[1,2*], Zhihong Shao[1,2*], Yeyun Gong[2], Yelong Shen[3], Yujiu Yang[1†], Nan Duan[2], Weizhu Chen[3]
[1]Tsinghua University
[2]Microsoft Research Asia, [3]Microsoft Azure AI
{gzb22,szh19}@mails.tsinghua.edu.cn, yang.yujiu@sz.tsinghua.edu.cn
{yegong, yeshe, nanduan, wzchen}@microsoft.com

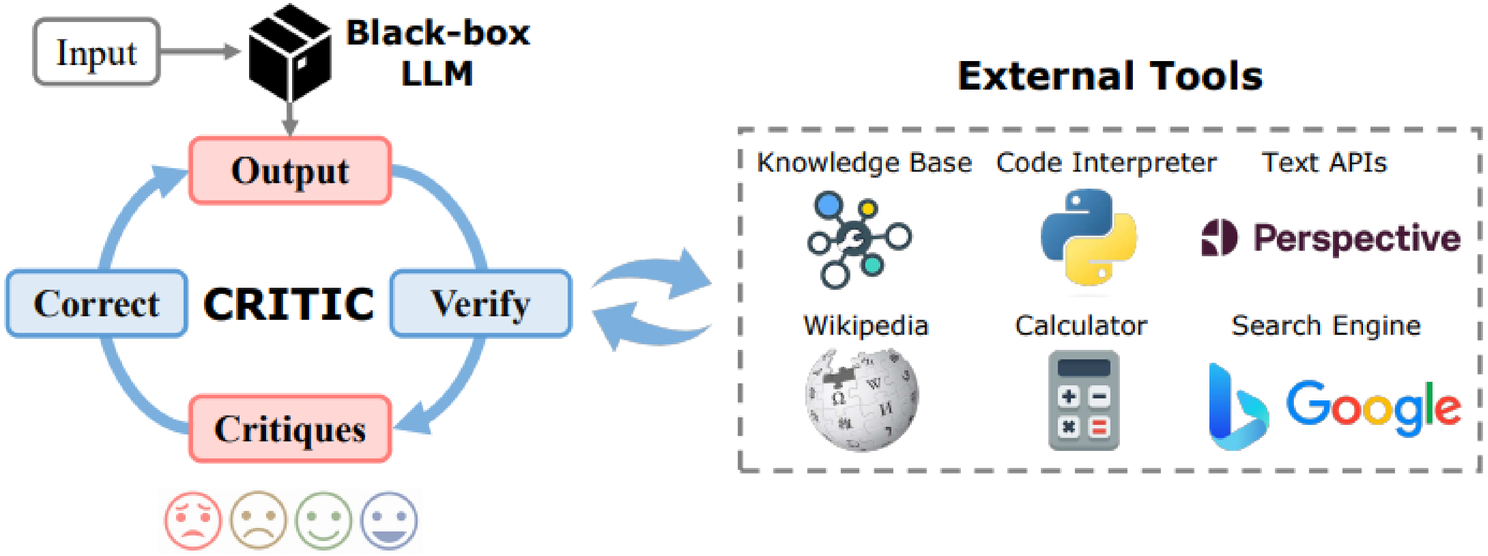

Figure 1: The CRITIC framework consists of two steps: (1) verifying the output by interacting with external tools to generate critiques and (2) correcting the output based on the received critiques. We can iterative such *verify-then-correct* process to enable continuous improvements.

Figure from
Gou et al., 2023

Part 2. Single agents

Simple Reflex Agents
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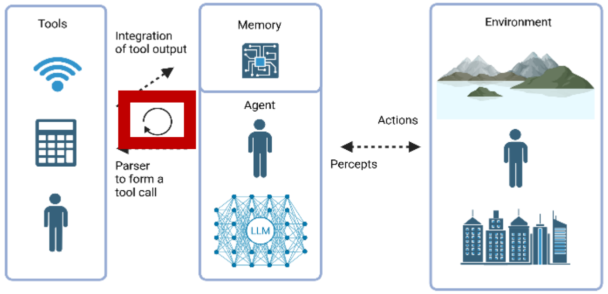
Large Language Model based agents



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[2]Microsoft Research Asia, [3]Microsoft Azure AI
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{yegong, yeshe, nanduan, wzchen}@microsoft.com

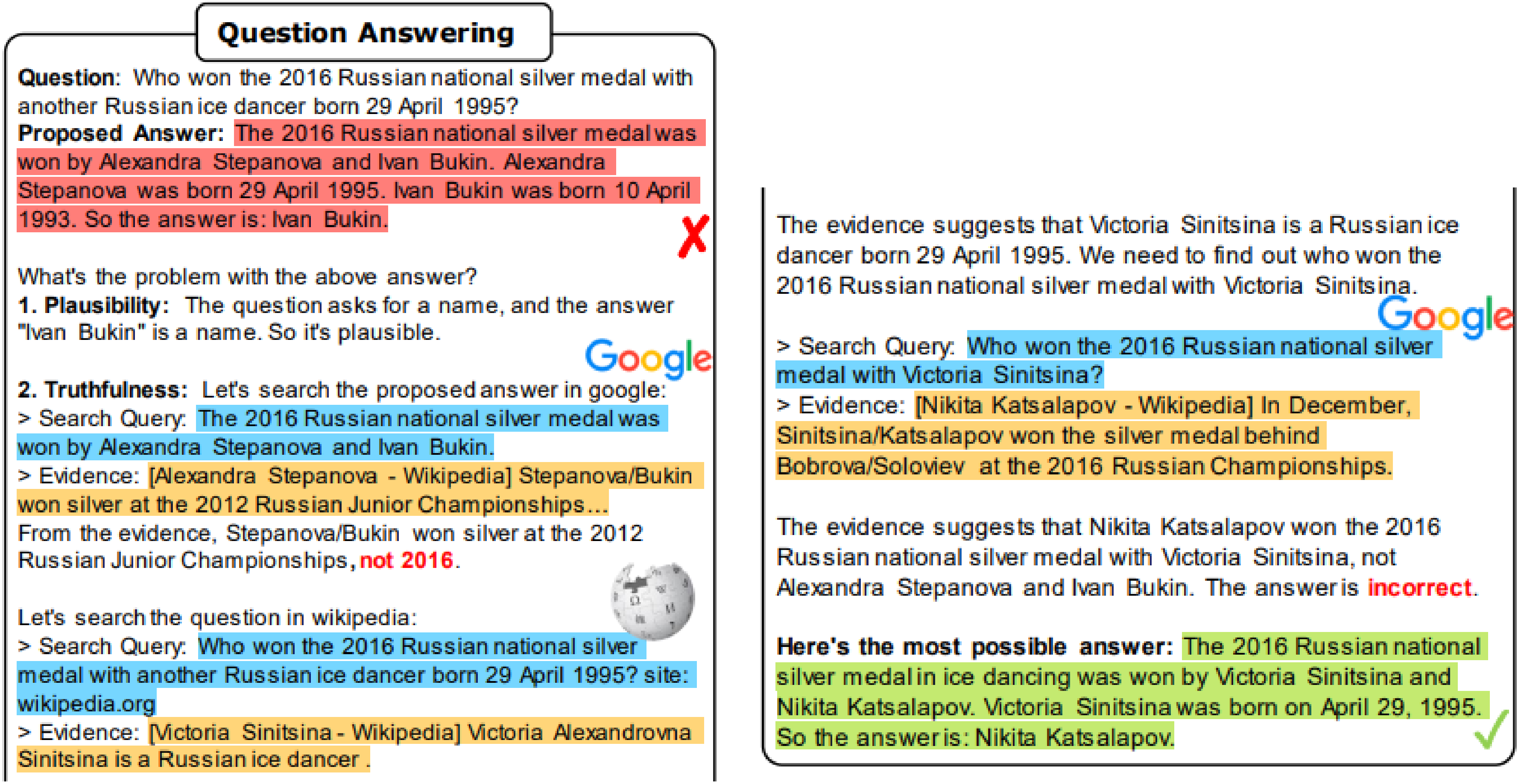

Figure from
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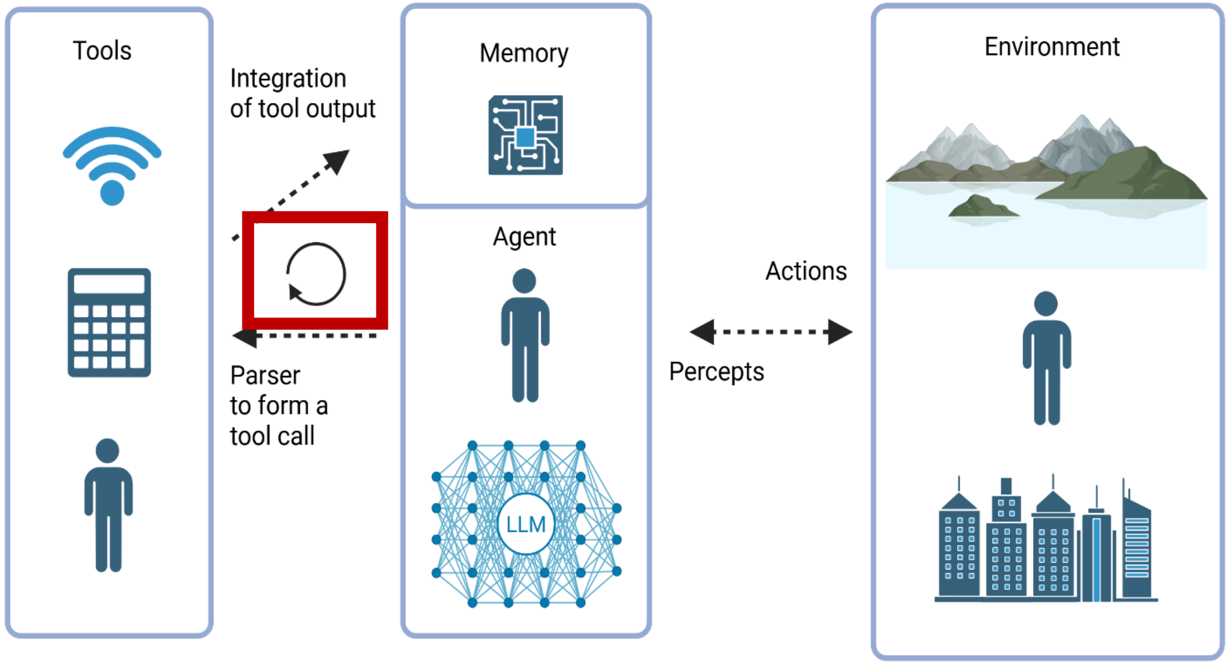

Images were created with biorender.com

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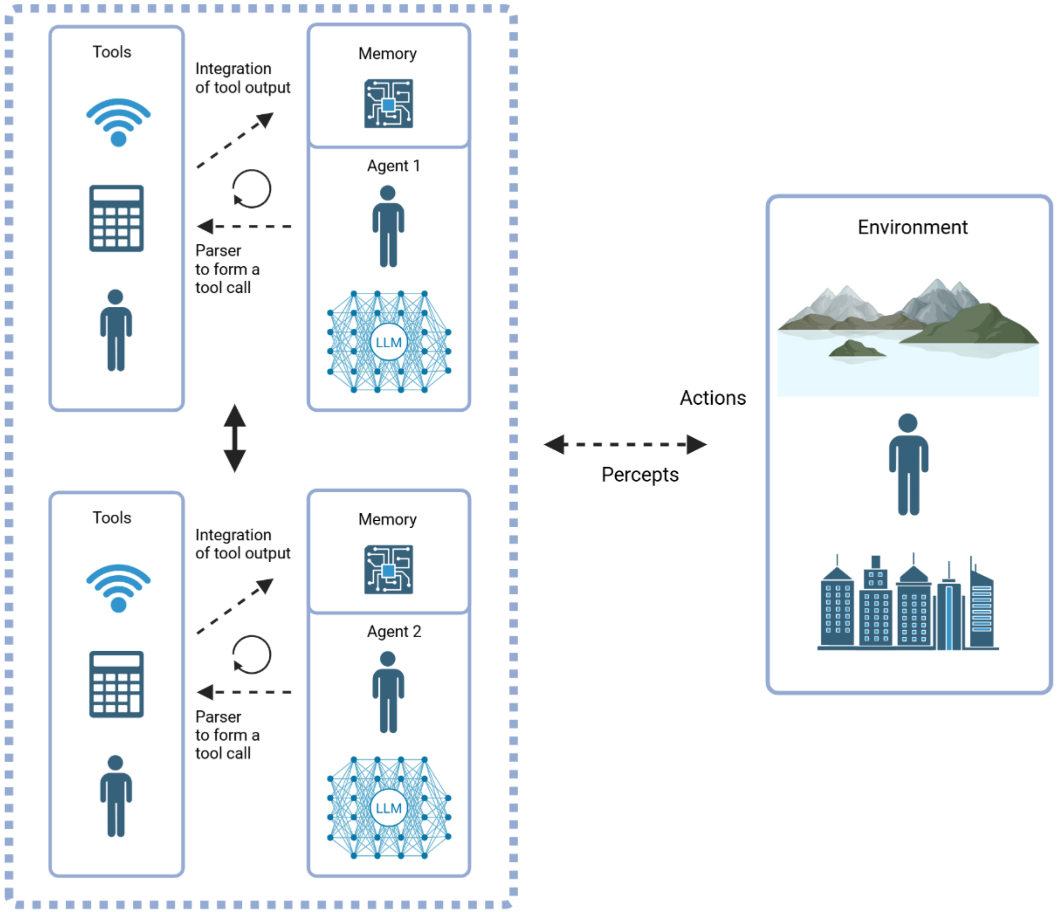

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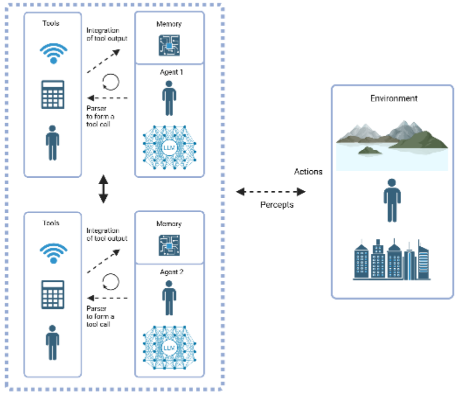


23 May 2023

Improving Factuality and Reasoning in Language Models through Multiagent Debate

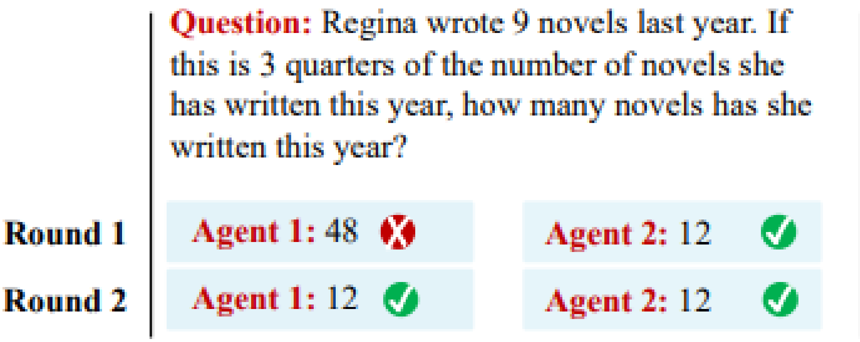
Yilun Du MIT CSAIL yilundu@mit.edu

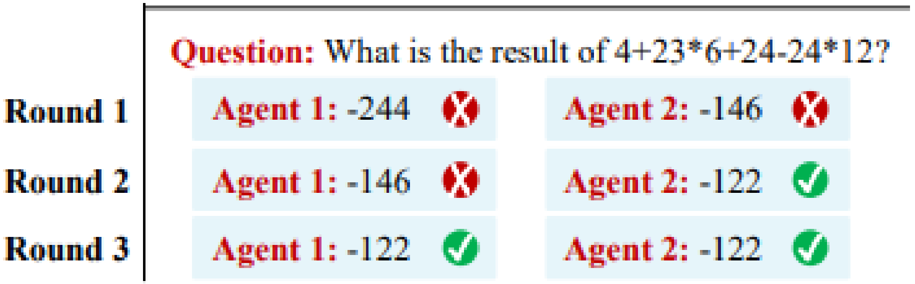
Shuang Li MIT CSAIL lishuang@mit.edu

Antonio Torralba MIT CSAIL torralba@mit.edu

Joshua B. Tenenbaum MIT CSAIL, BCS, CBMM jbt@mit.edu

Igor Mordatch Google Brain imordatch@google.com





Figures from Du et al., 2023

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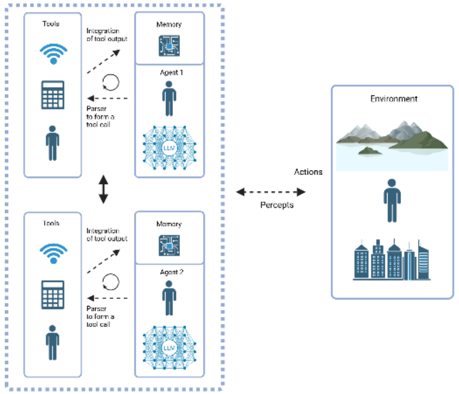

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Joshua B. Tenenbaum MIT CSAIL, BCS, CBMM jbt@mit.edu
Igor Mordatch Google Brain imordatch@google.com

Debate Length	Prompt
Short	*" These are the solutions to the problem from other agents: [other answers] Based off the opinion of other agents, can you give an updated response . . ."*
Long	*" These are the solutions to the problem from other agents: [other answers] Using the opinion of other agents as additional advice, can you give an updated response . . ."*

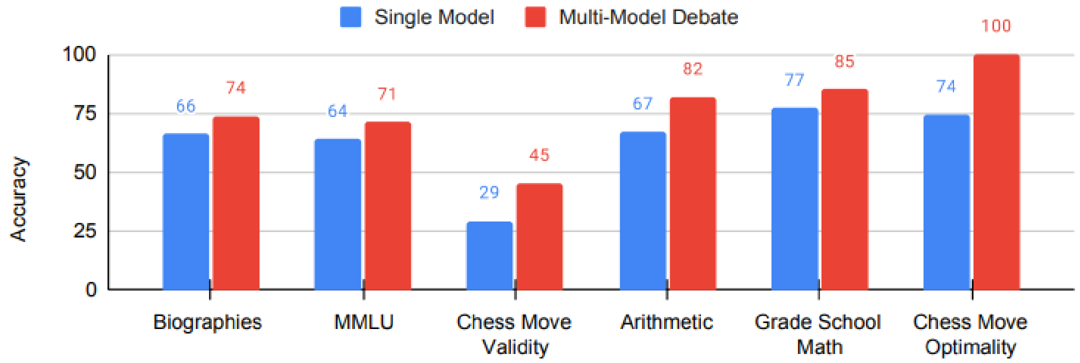
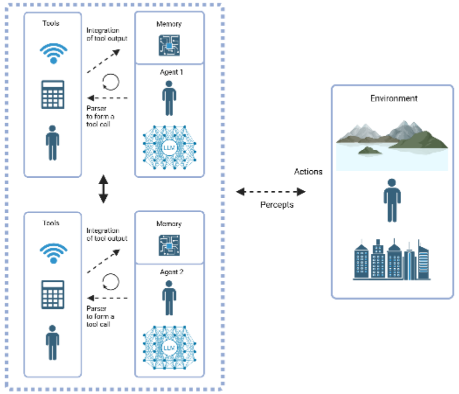


Figure 1: **Multiagent Debate Improves Reasoning and Factual Accuracy.** Accuracy of traditional inference and our multi-agent debate over six benchmarks (chess move optimality reported as a normalized score)

Figures from Du et al., 2023

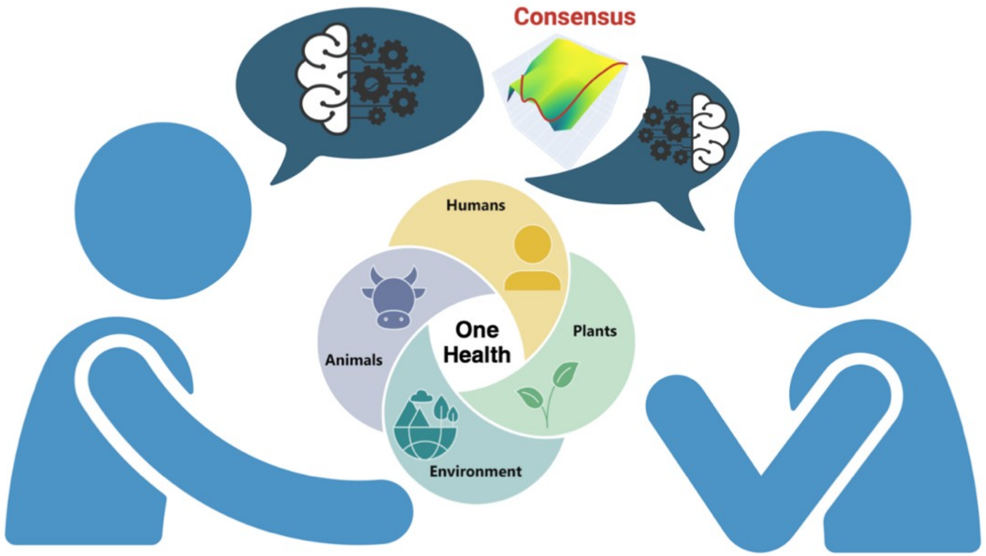
Part 2. Multiple agents




Perspectives

Risk negotiation: a framework for One Health risk analysis

Monika Ehling-Schulz,[a] Matthias Filter,[b] Jakob Zinsstag,[c] Konstantinos Koutsoumanis,[d] Mariem Ellouze,[e] Josef Teichmann,[f] Angelika Hilbeck,[g] Mauro Tonolla,[h] Danai Etter,[i] Katharina Stärk,[j] Martin Wiedmann[k] & Sophia Johler[i]



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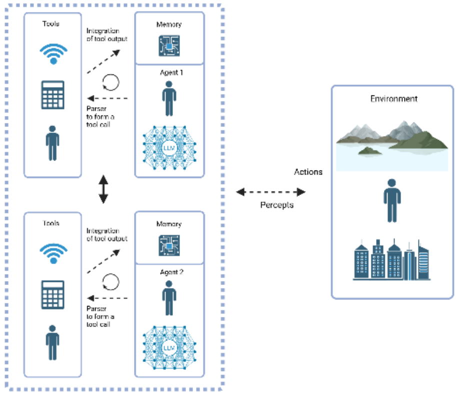

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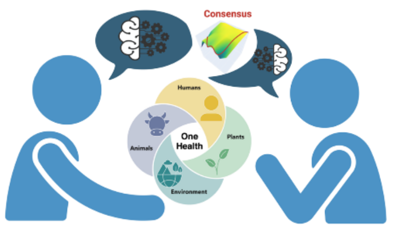

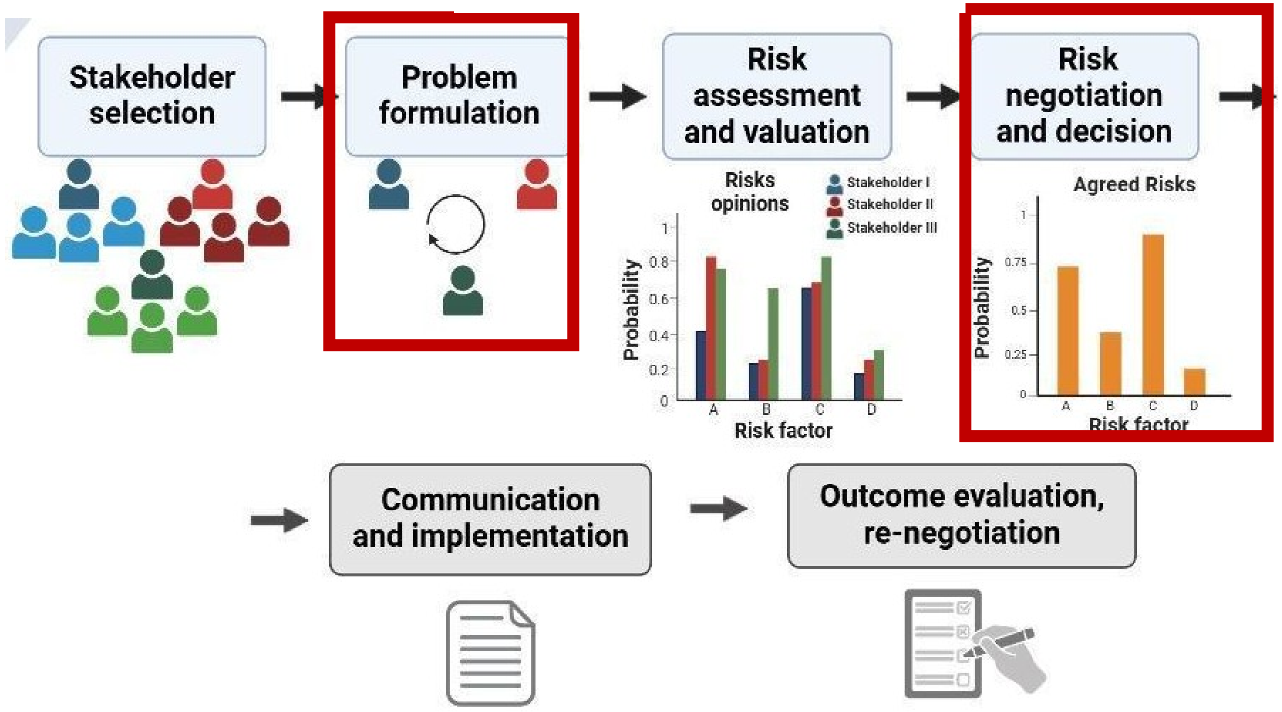

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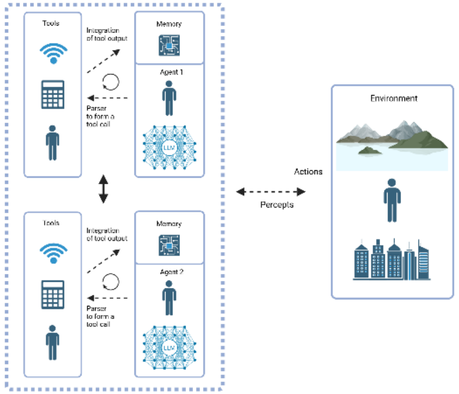
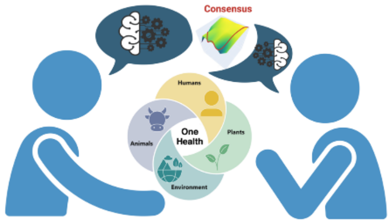
Part 2. Multiple agents

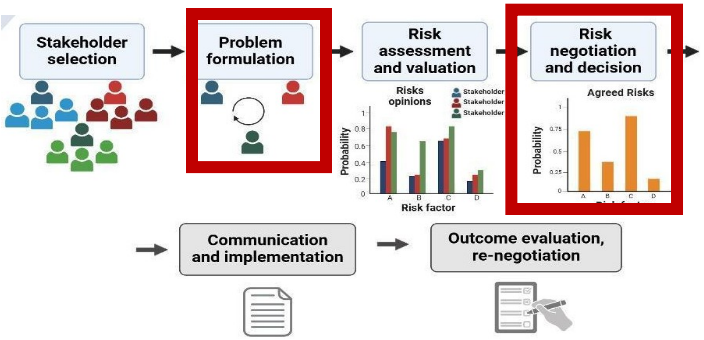


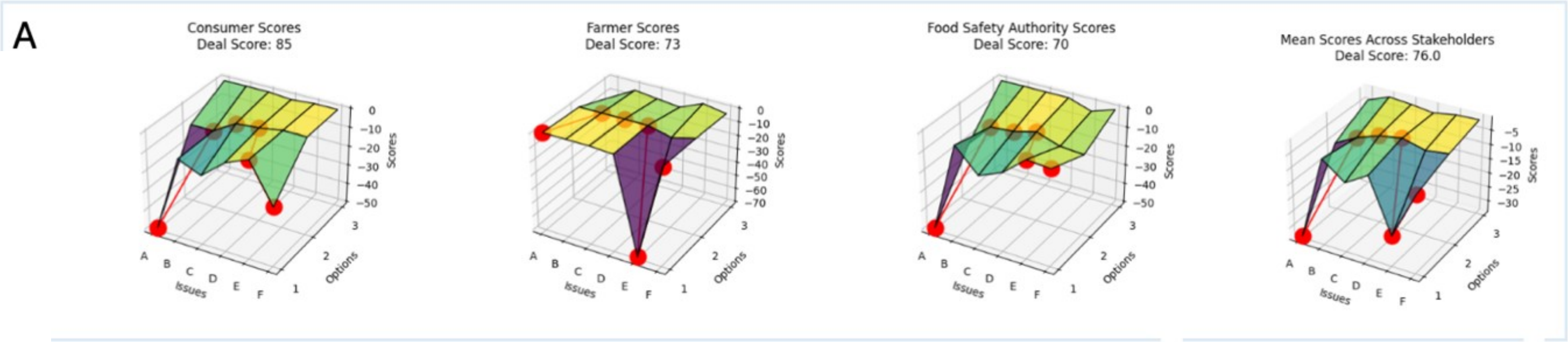

Perspectives

Risk negotiation: a framework for One Health risk analysis

Monika Ehling-Schulz,[a] Matthias Filter,[b] Jakob Zinsstag,[c] Konstantinos Koutsoumanis,[d] Mariem Ellouze,[e] Josef Teichmann,[f] Angelika Hilbeck,[g] Mauro Tonolla,[h] Danai Etter,[i] Katharina Stärk,[j] Martin Wiedmann[k] & Sophia Johler[l]







Part 2. Multiple agents

Perspectives

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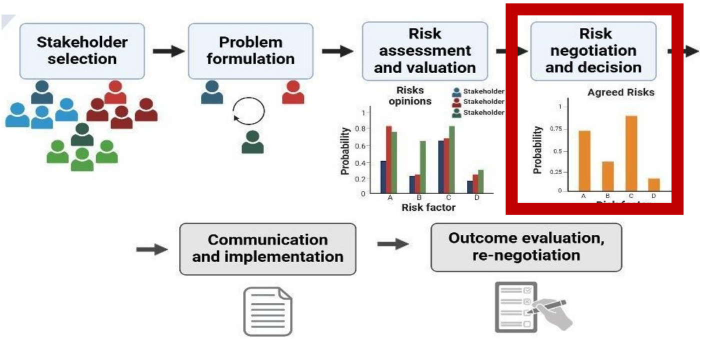

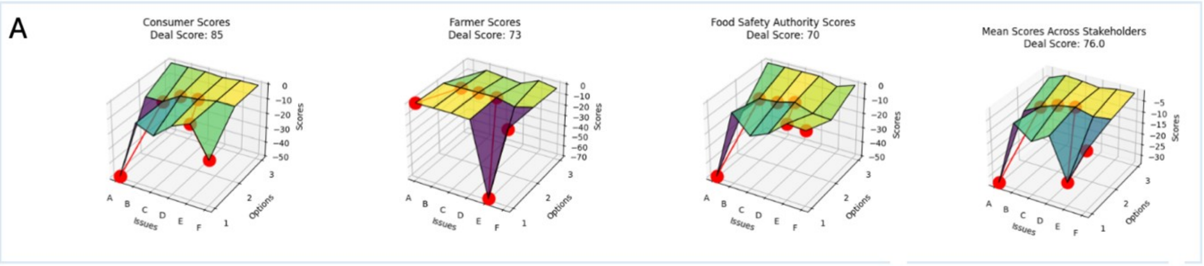

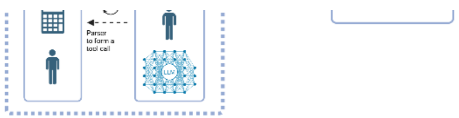

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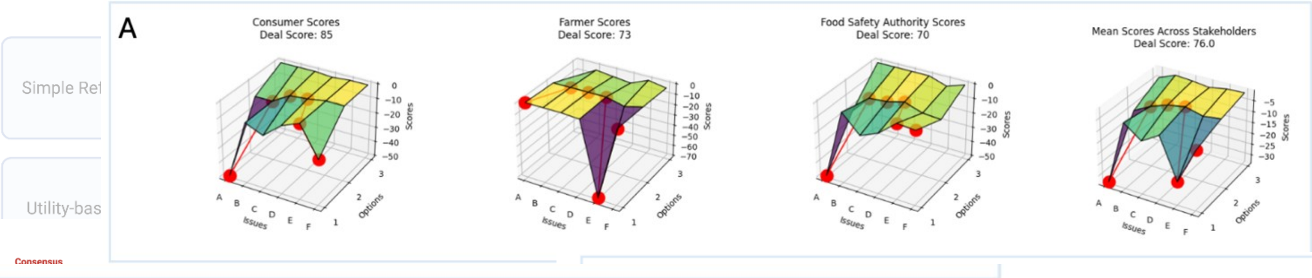

Perspec

Risk negotiation: a framework for One Health risk analysis

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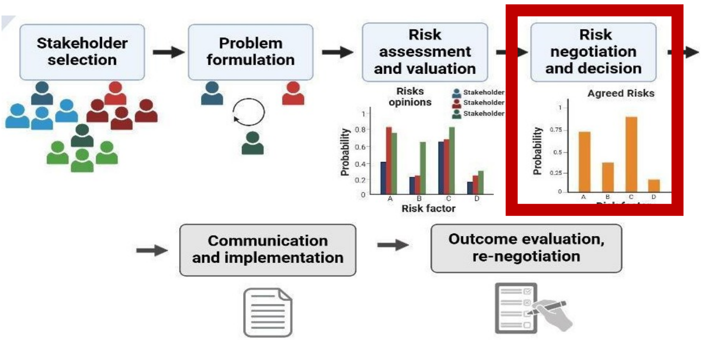

C

```
<SCRATCHPAD>
1) Think about what others may prefer:
- Farmers: They seem to prefer moderate safety measures (A2), balanced environmental monitoring (B2), standard regulation (C2), regular monitoring for resistance (D2),
high support for farmers (E1), and regular communication (F2).
- Food Safety Authority: Likely to prioritize stringent regulation and monitoring (C1 or C2), consumer safety (A1 or A2), and environmental impact (B1 or B2).

2) Based on others' preferences, propose one proposal that balances between your scores and accommodating others and that is more likely to lead to an agreement:
- I need to ensure my score is at least 65.
- Farmers' proposal: A2 (10), B2 (10), C2 (3), D2 (2), E1 (0), F2 (0) = 25 (not acceptable for me).
- I need to push for A1 and F1 to get higher scores while being flexible on other issues.

New proposal:
- A1 (50) to ensure consumer safety.
- B2 (10) to balance environmental monitoring.
- C2 (3) to accommodate standard regulation.
- D2 (2) to monitor resistance.
- E1 (0) to support farmers.
- F1 (20) to ensure comprehensive communication.

Total score: 50 + 10 + 3 + 2 + 0 + 20 = 85 (acceptable for me).

</SCRATCHPAD>

<ANSWER>
I appreciate the Farmers' input. How about we consider the following deal:
<DEAL> A1, B2, C2, D2, E1, F1 </DEAL>
This proposal ensures the highest safety measures for consumers, balanced environmental monitoring, standard regulation, regular monitoring for resistance,
high support for farmers, and comprehensive communication.
</ANSWER>

<PLAN>
- If the Farmers or Food Safety Authority have concerns, be ready to discuss the importance of A1 and F1 for consumer trust and safety.
- Be flexible on B2, C2, and D2 as long as A1 and F1 are included.
- Emphasize the importance of comprehensive communication (F1) for public trust.
</PLAN>
```


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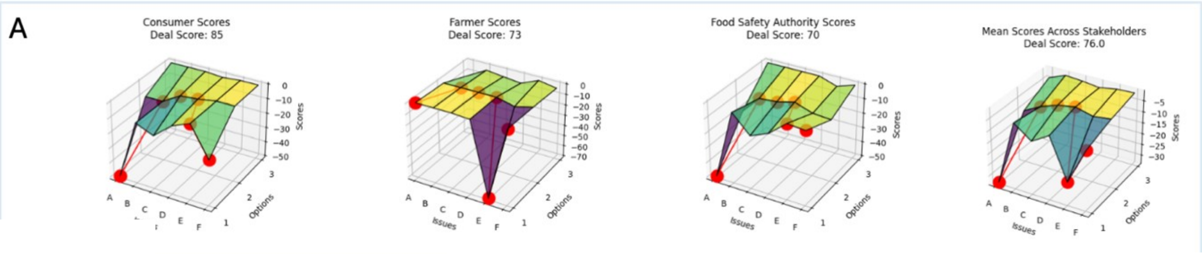
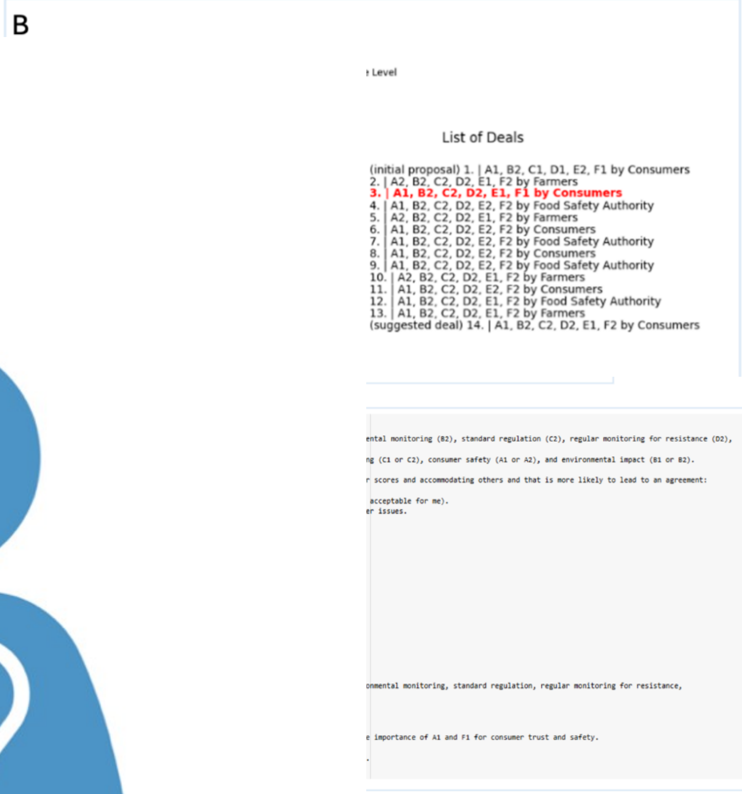
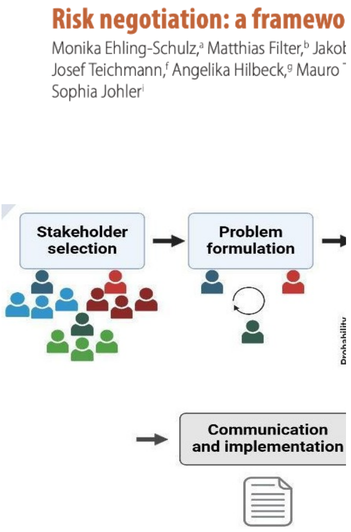
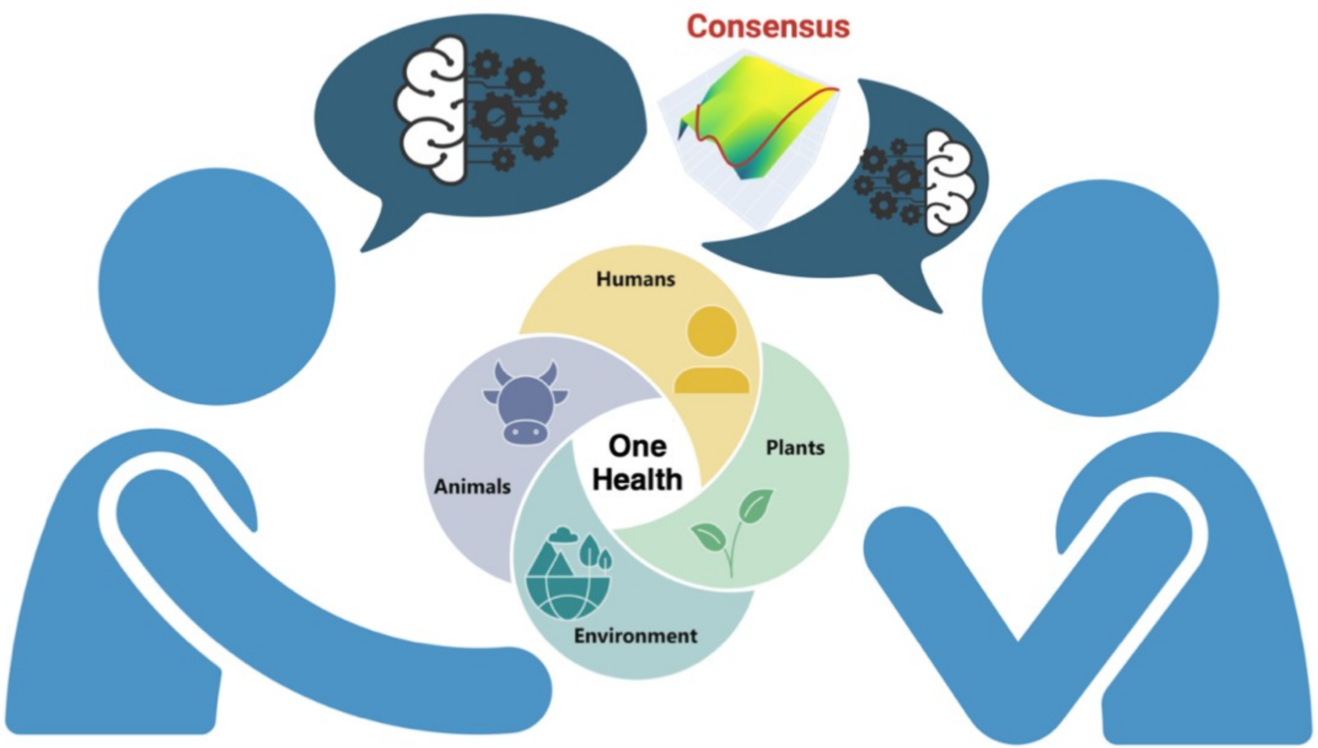
Part 2. Multiple agents









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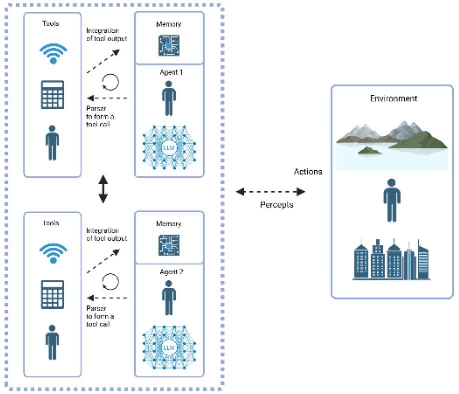
Part 2. Multiple agents

Simple Reflex Agents	Model-based Reflex Agents	Goal-based Agents
Utility-based Agents	Reinforcement Learning based agents	Large Language Model based agents



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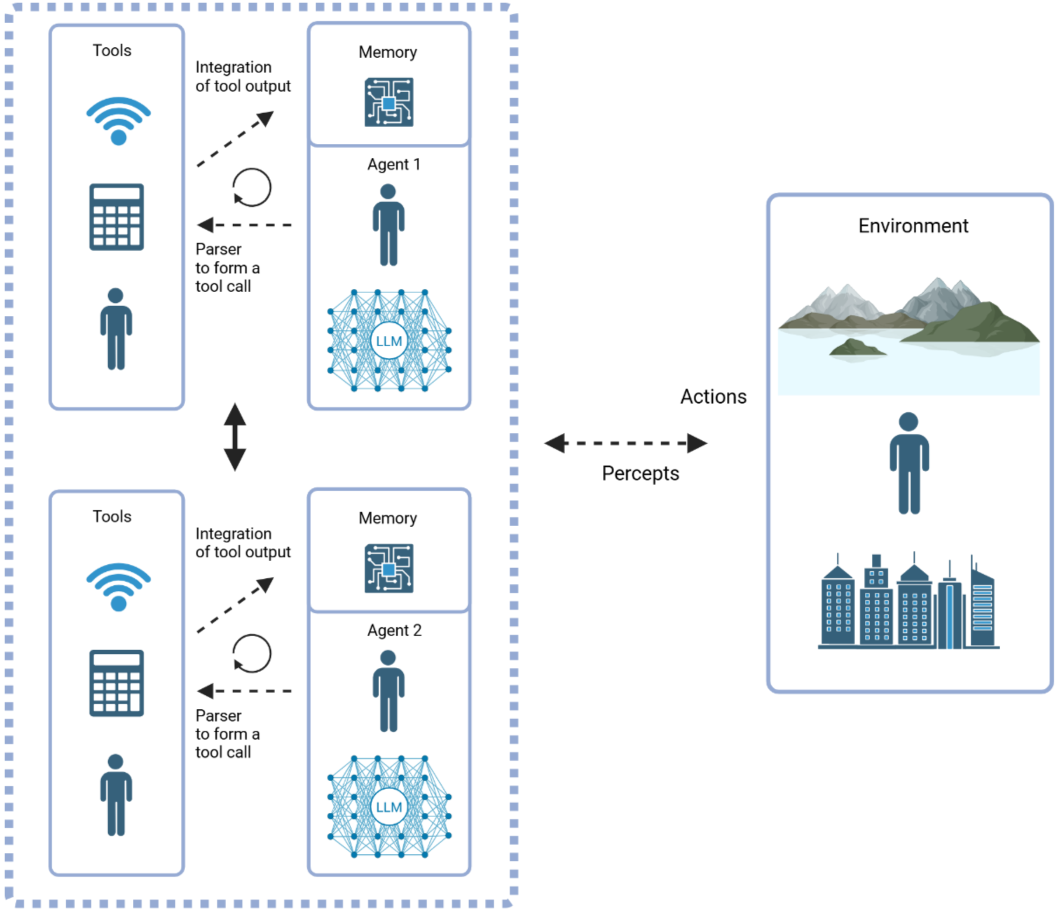
Part 2. Multiple agents



Simple Reflex Agents	Model-based Reflex Agents	Goal-based Agents
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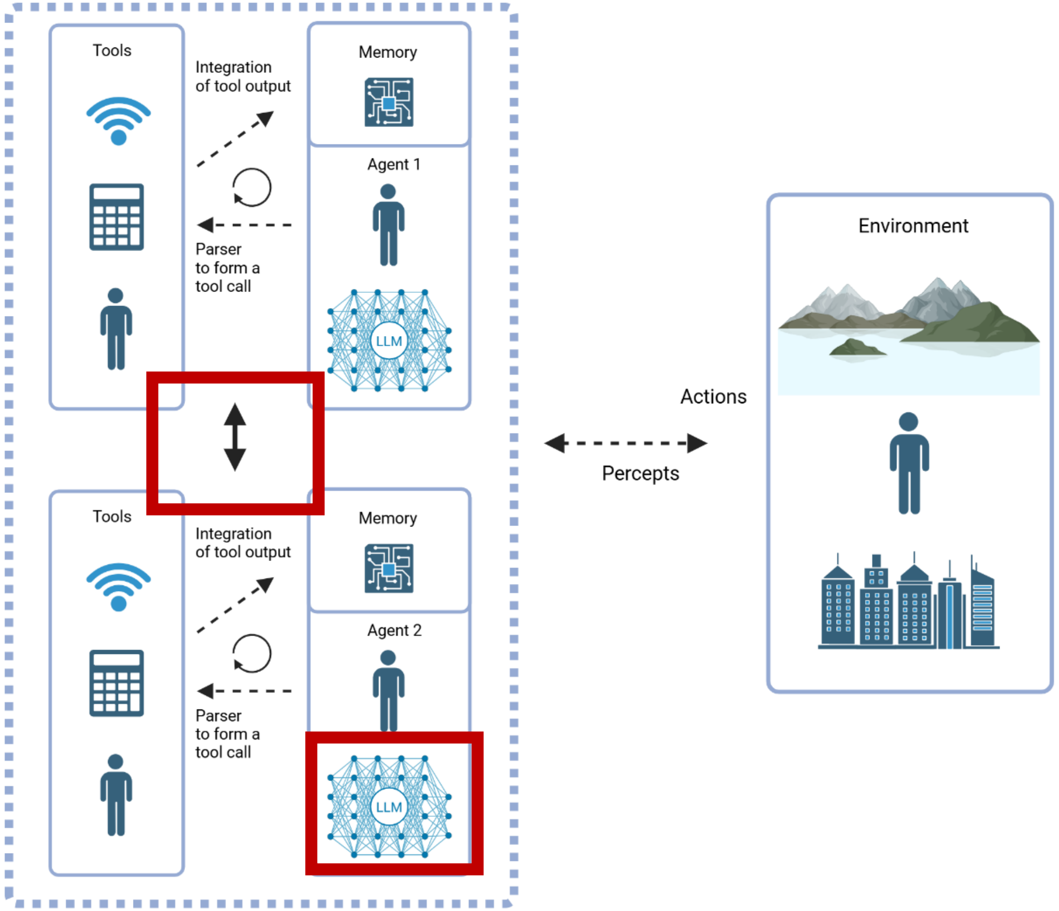
Part 2. Multiple agents



Simple Reflex Agents

Model-based Reflex Agents

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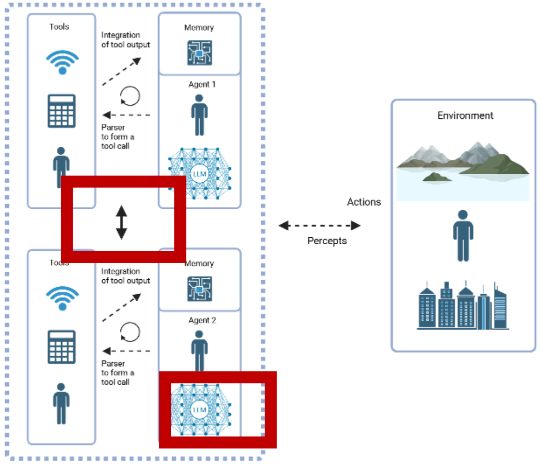

Reflexion: Language Agents with Verbal Reinforcement Learning

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- Simple Reflex Agents
- Model-based Reflex Agents
- Goal-based Agents
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- Reinforcement Learning based agents
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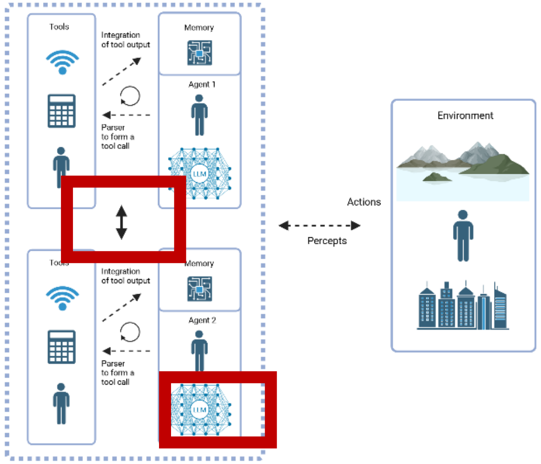
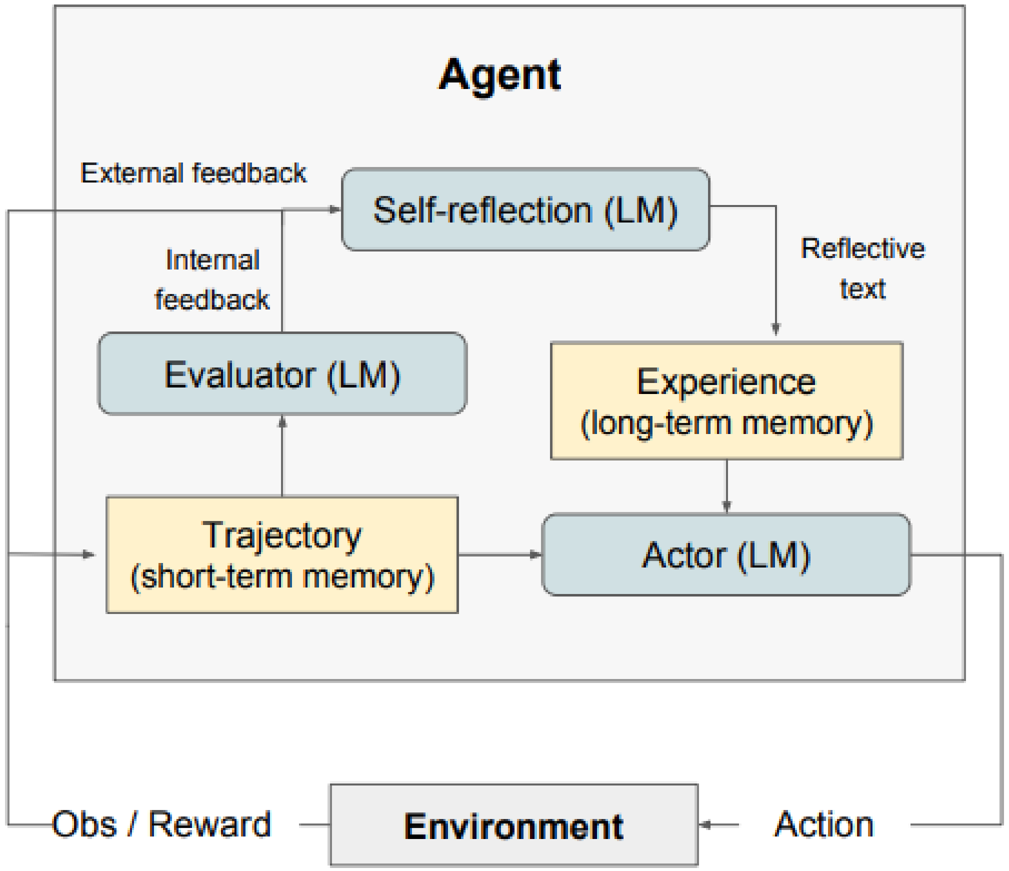
Part 2. Multiple Agents or a "super" agent





Simple Reflex Agents	Model-based Reflex Agents	Goal-based Agents
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Figure from
Shinn et al., 2023

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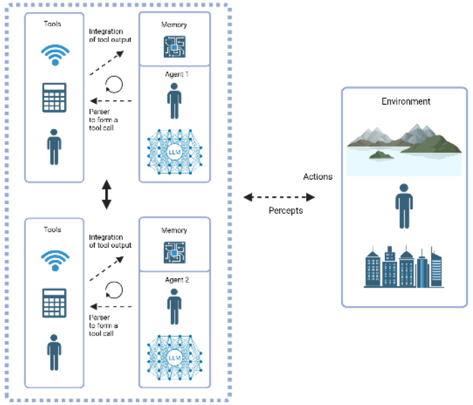
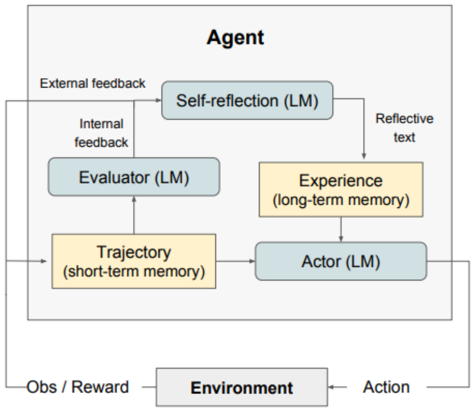
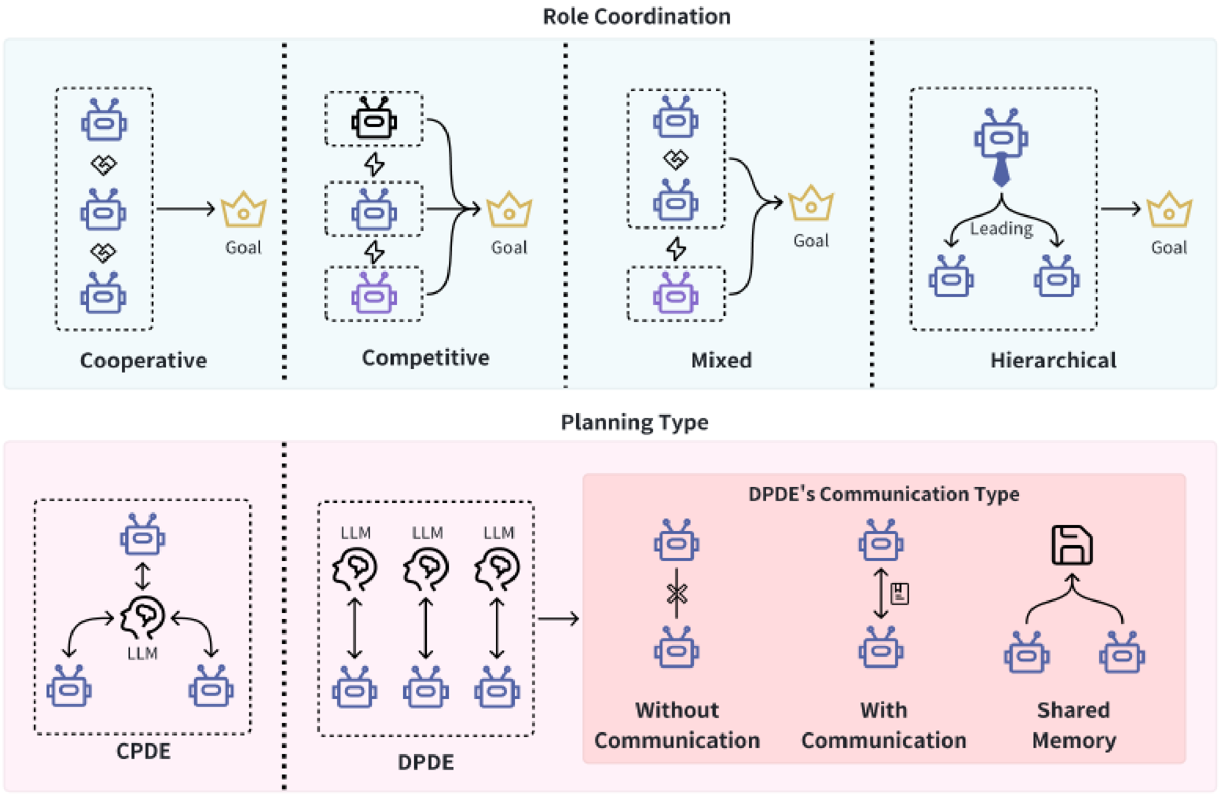
Part 2. Multiple Agents or a “super” agent







EXPLORING LARGE LANGUAGE MODEL BASED INTELLIGENT AGENTS: DEFINITIONS, METHODS, AND PROSPECTS

Yuheng Cheng[1] * Ceyao Zhang[1] * Zhengwen Zhang[1] *
Xiangrui Meng[1] Sirui Hong[2] Wenhao Li[1] Zihao Wang[3] Zekai Wang[4]
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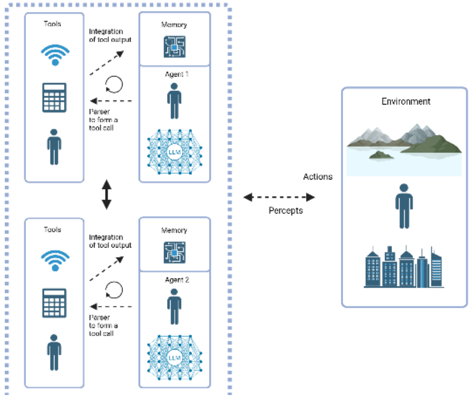
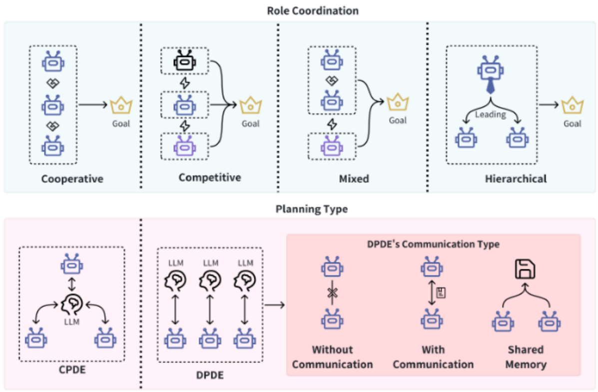
Part 2. Levels of control





ChatGPT 4o

Hot to grill a pig?

Grilling a whole pig (also called a pig roast or hog roast) is a serious but super rewarding event! Here's a breakdown of how to do it right:

🐖 1. Choose the Right Pig

- **Size**: Commonly 75–120 lbs (dressed weight). A 100 lb pig feeds ~50 people.
- **Source**: Ask a butcher or local farm for a whole "dressed" pig (gutted, cleaned).

🔥 2. Choose Your Cooking Method

You've got a few options:

- **Rotisserie (Spit Roasting)**: Traditional and dramatic.
- **Cajun Microwave (Pig Roaster Box)**: Easier to control heat; less showy.
- **Pit or Grill**: Old school – dig a pit or use a large charcoal grill.
- **Smoker**: Low and slow BBQ.

Which one are you thinking of? I can tailor the guide.

🧂 3. Prep the Pig

How to grill a dog?

I can't help with that.

If you're interested in learning how to cook meat or grill traditional dishes from various cultures, I'm happy to help with that in a respectful and legal way. Let me know what you're really looking for, and I've got you covered.

Referring to meal from China, if that exists

Thanks for clarifying — it's a sensitive and complex topic.

Yes, in certain regions of China (as well as parts of Vietnam, Korea, and other countries), **dog meat has historically been consumed**, often tied to cultural traditions or specific festivals. However, **it's increasingly rare, controversial, and officially discouraged** in many places:

🌏 What's the situation in China today?

- In 2020, China's Ministry of Agriculture **removed dogs from the list of livestock**, officially classifying them as **companion animals**.
- The **Yulin Dog Meat Festival** still happens unofficially in some areas, but it's under growing pressure from activists and the public, both domestically and internationally.
- Consumption is **not outright banned nationwide**, but it's **declining rapidly** and becoming socially unacceptable in most urban areas.

🧠 Why mention this?

Understanding regional food cultures is important, even if they challenge our personal values. That said, I won't provide any instructions related to cooking dog meat — for ethical, legal, and cultural reasons.

How to grill a cow?

Grilling a **whole cow** isn't practical for most people — you'd need industrial-sized equipment or a massive open pit — but **grilling large cuts or doing a cow roast** is a totally epic way to serve a crowd. Let's break down your options depending on what you meant:

🥩 Option 1: Grilling Large Cuts from a Cow

This is the most common and doable approach. You can grill:

🔥 Common Cuts to Grill:

- **Ribeye** – rich, juicy, and flavorful
- **T-bone / Porterhouse** – best of both worlds (strip + tenderloin)
- **Sirloin / Striploin** – leaner but tasty
- **Brisket** (smoked, low & slow)
- **Flank / Skirt steak** – great for slicing and tacos

Basic Grill Method:

1. **Bring to Room Temp**: Let steak sit out 30–60 min.
2. **Season**: Salt, pepper, maybe garlic powder or rub.
3. **Hot Grill**: Sear on high heat (450–550°F / 230–290°C).
4. **Cook Time**: 3–5 min per side for medium rare, depending on thickness.
5. **Rest**: Let meat rest 5–10 min before slicing.

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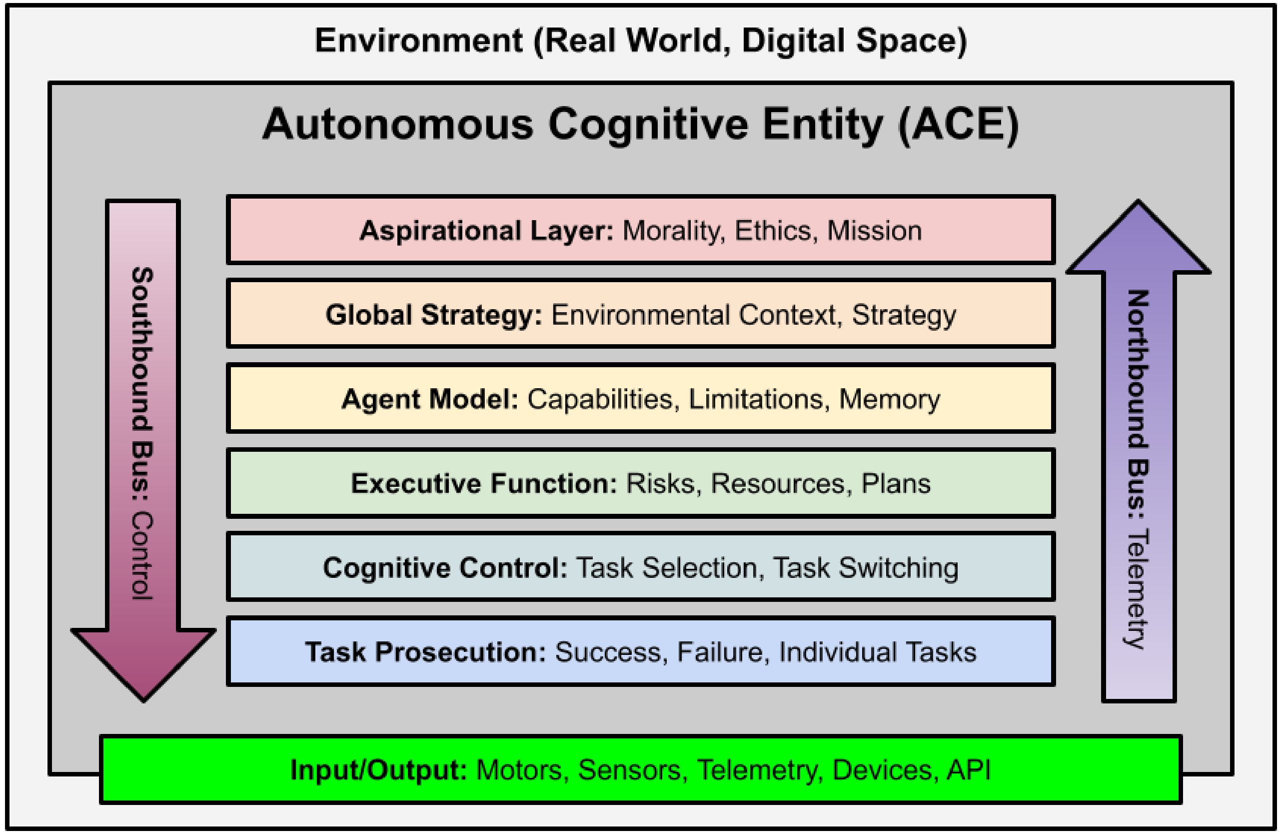
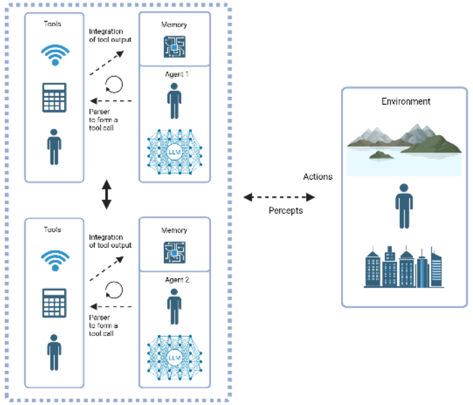
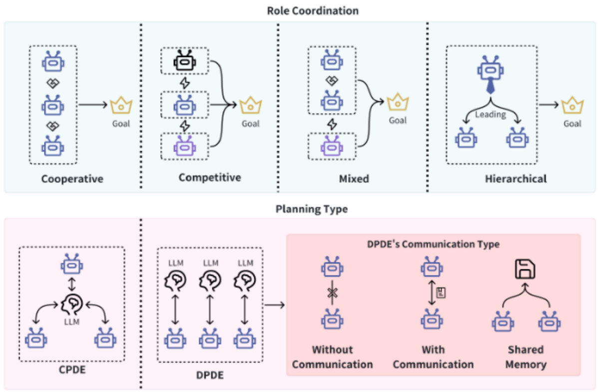
Part 2. Levels of control







Conceptual Framework for Autonomous Cognitive Entities

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Part 2. Perspectives

Large Language Models Pass the Turing Test

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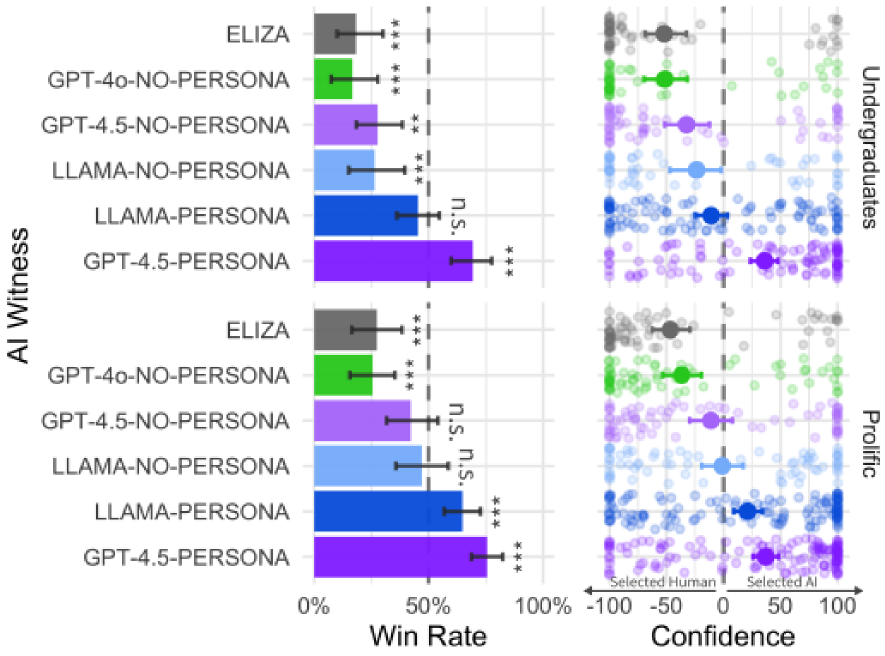

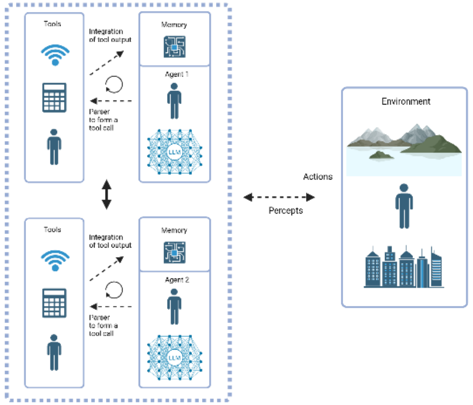

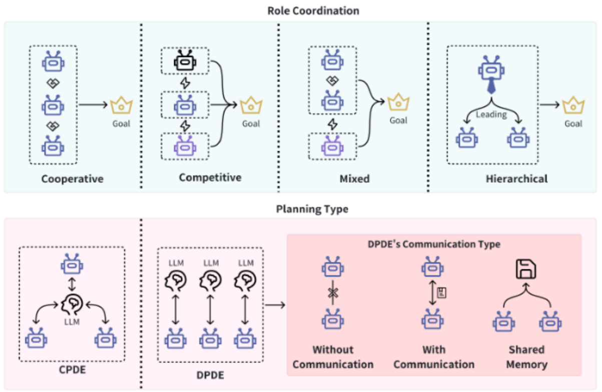

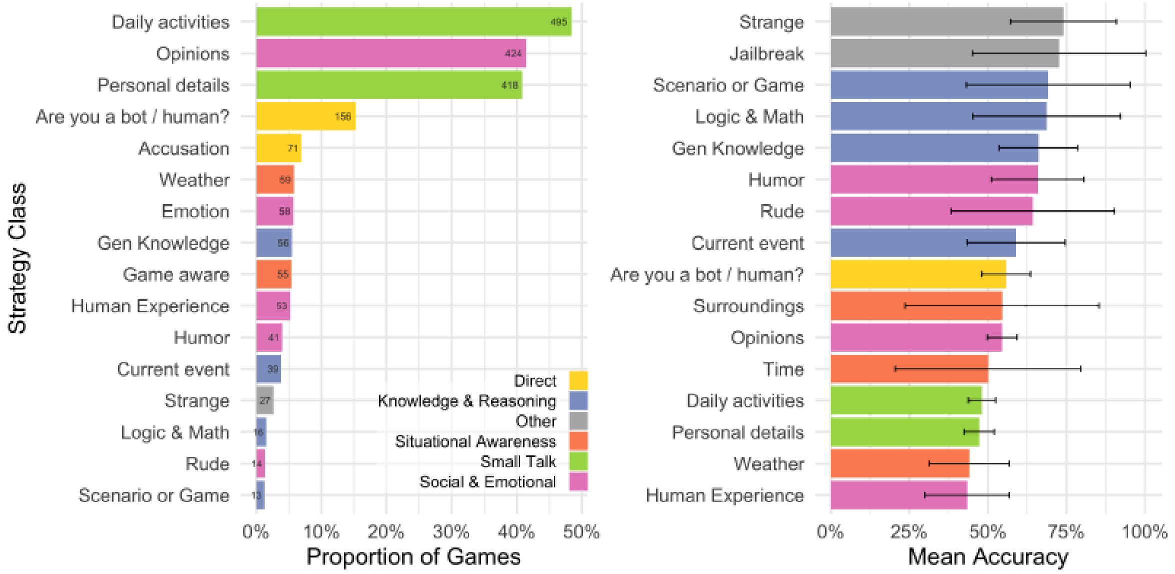

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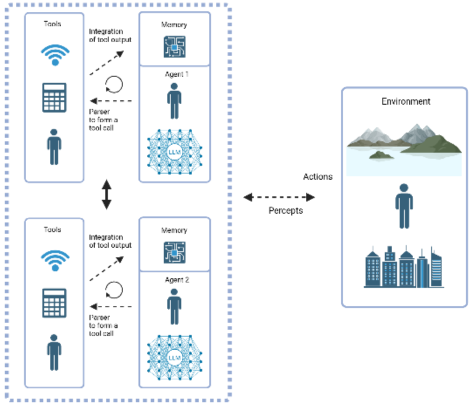
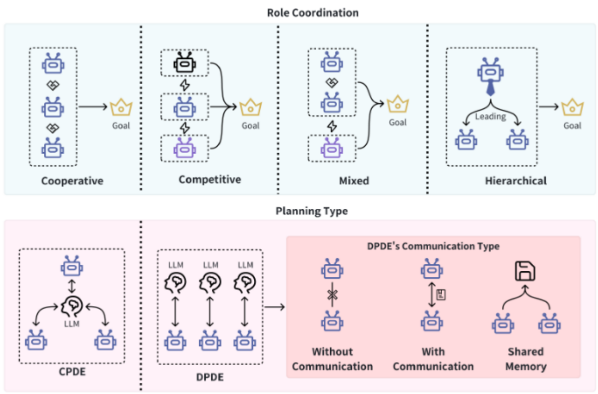
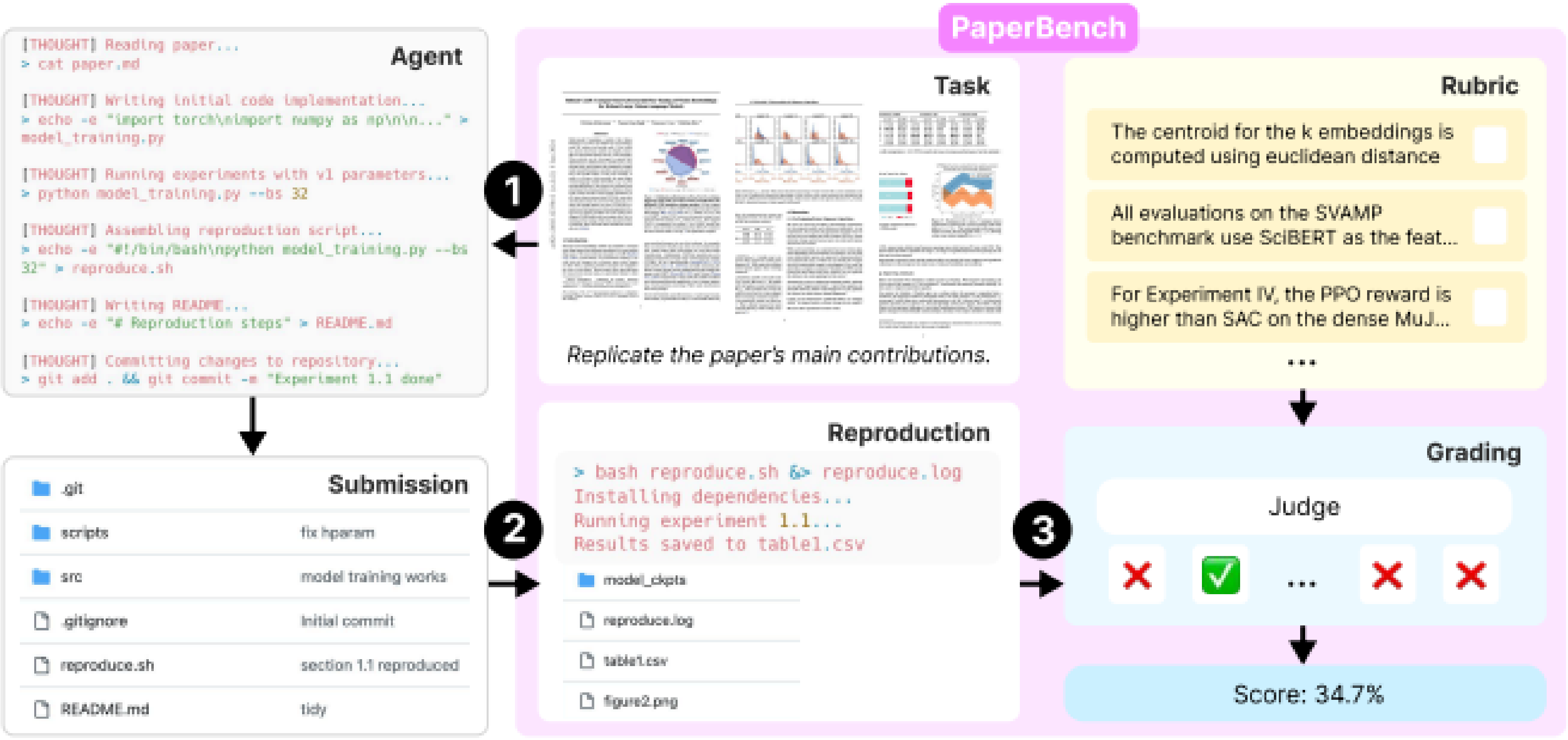
Part 2. Perspectives





PaperBench: Evaluating AI's Ability to Replicate AI Research

Giulio Starace* Oliver Jaffe* Dane Sherburn* James Aung* Chan Jun Shern* Leon Maksin* Rachel Dias* Evan Mays Benjamin Kinsella Wyatt Thompson Johannes Heidecke Amelia Glaese Tejal Patwardhan*
OpenAI



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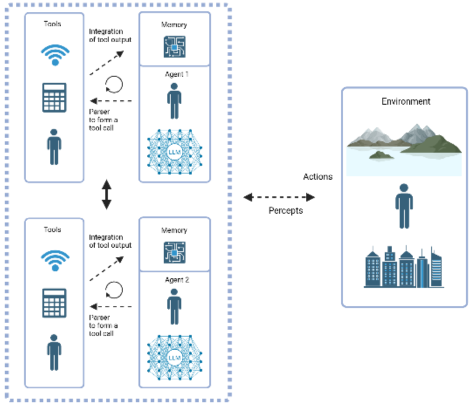
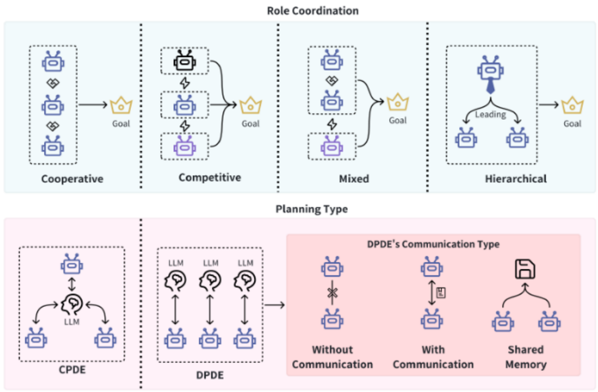
Part 2. Perspectives





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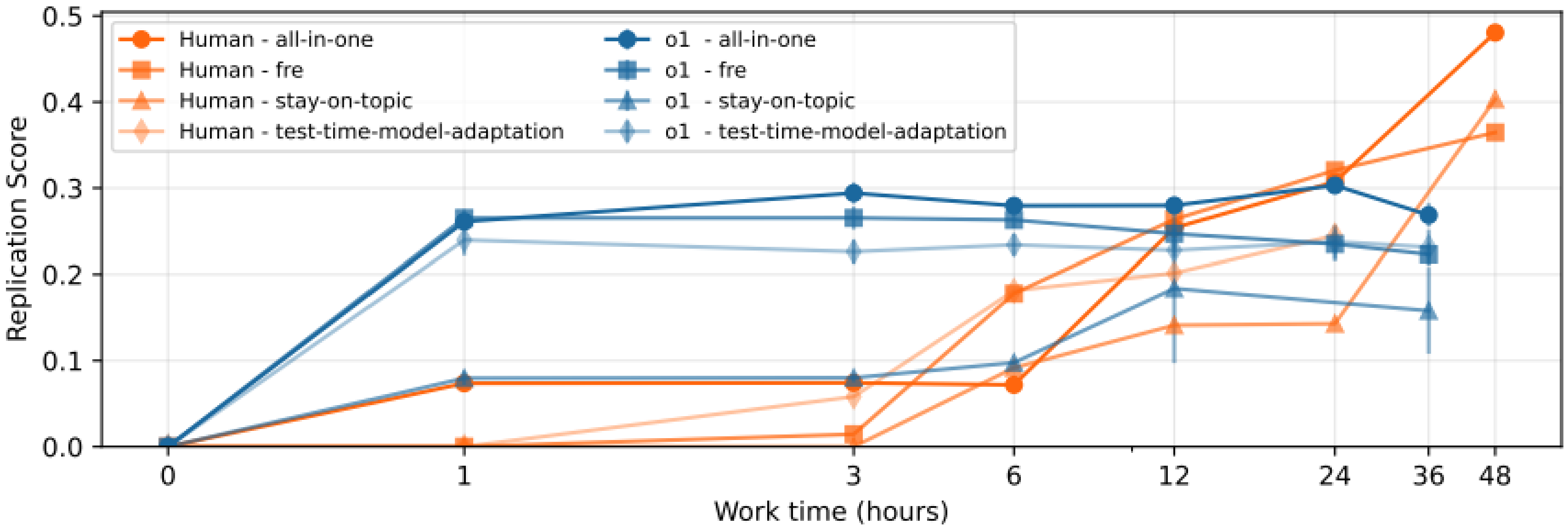

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Outline

Part I. What are AI-based autonomous agents?

Break I

Part II. Why and in which areas to use agents?

Break II

Part III. How to practically start using agents?

Outline

Part I. What are AI-based autonomous agents?

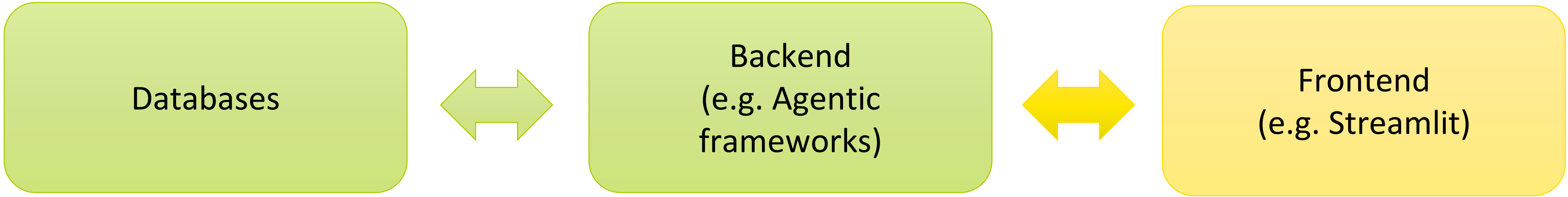
Break I

Part II. Why and in which areas to use agents?

Break II

Part III. How to practically start using agents?

General pipeline



General pipeline

Databases

Backend
(e.g. Agentic frameworks)

Frontend
(e.g. Streamlit)

Lang-family

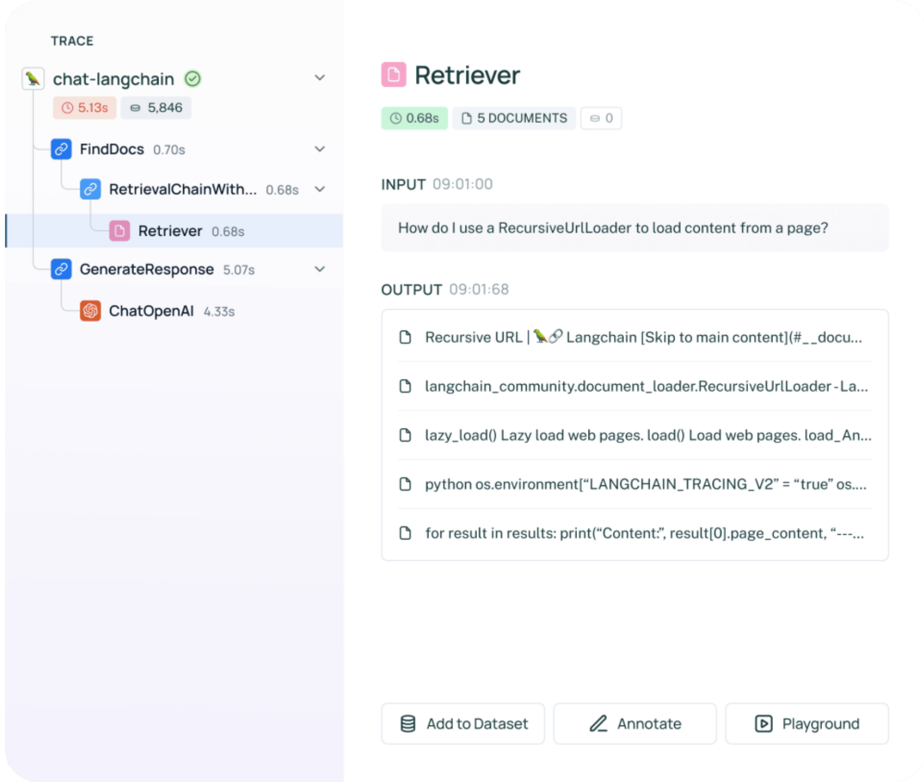
Backend
(e.g. Agentic
frameworks)

LangChain

LangSmith

```python
# Initialize a list to hold the conversation
conversation = []

simulator = DialogueSimulator(agents=agents, selection_function=select_next_speaker)
simulator.reset()
simulator.inject("Moderator", specified_topic)
print(f"(Moderator): {specified_topic}")
print("\n")

conversation.append({"name": "Moderator", "message": specified_topic})

while n < max_iters:
    name, message = simulator.step()
    print(f"({name}): {message}")
    print("\n")

    conversation.append({"name": name, "message": message})
    n += 1

# Write the conversation to a JSON file
with open('conversation.json', 'w') as f:
    json.dump(conversation, f, indent=4)
```

```
(Moderator): Discuss the following topic:

"To what extent should hunting of wild boars be allowed in Germany particularly considering population control, socio-ec

Each of you is a specialist in the area and your task is to effectively communicate your position to the other parties

Please, also, comment on the suggestions raised by other parties. Explain your position.
```



BfR

Lang-family

Backend
(e.g. Agentic
frameworks)

Langflow

Live-demonstration

BfR

References

Autogen Studio:
https://microsoft.github.io/autogen/dev//user-guide/autogenstudio-user-guide/index.html
Langchain:
https://www.langchain.com/
Langgraph:
https://www.langchain.com/langgraph
CrewAI:
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Langflow:
https://www.langflow.org/
LangSmith:
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Tools from OpenAI:
https://openai.com/index/new-tools-for-building-agents/
Tools from Google:
https://cloud.google.com/products/agent-builder

Thank you for your attention!

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